

Deubiquitylase OTUD3 regulates PTEN stability and suppresses tumorigenesis

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PTEN is one of the most frequently mutated tumour suppressors and reduction in PTEN protein stability also plays a role in tumorigenesis. Although several ubiquitin ligases for PTEN have been identified, the deubiquitylase for de-polyubiquitylation and stabilization of PTEN is less defined. Here, we report OTUD3 as a deubiquitylase of PTEN. OTUD3 interacts with, de-polyubiquitylates and stabilizes PTEN. Depletion of OTUD3 leads to the activation of Akt signalling, induction of cellular transformation and cancer metastasis. OTUD3 transgenic mice exhibit higher levels of the PTEN protein and are less prone to tumorigenesis. Reduction of OTUD3 expression, concomitant with decreased PTEN abundance, correlates with human breast cancer progression. Furthermore, we identified loss-of-function OTUD3 mutations in human cancers, which either abolish OTUD3 catalytic activity or attenuate the interaction with PTEN. These findings demonstrate that OTUD3 is an essential regulator of PTEN and that the OTUD3–PTEN signalling axis plays a critical role in tumour suppression.

PTEN (phosphatase and tension homologue deleted on chromosome 10) has been identified as one of the most frequently mutated genes in human cancers^{1,2} as well as inherited syndromes^{3,4}. Heterozygous deletion of *Pten* leads to multiple tumours in mice, whereas homozygous mice die as embryos^{5–7}. PTEN is a phosphatase that dephosphorylates PtdIns(3,4,5)P₃ and inhibits the PI(3)K/Akt signalling in the cytoplasm^{8,9}. PTEN also has phosphatase-independent activities and PI(3)K/Akt-independent functions in the nucleus^{10,11}. PTEN activity, expression and localization are regulated by post-translational modifications, including ubiquitylation⁹. Several E3 ubiquitin ligases including Nedd4-1, WWP2, CHIP, XIAP and TRIM27 have been reported to promote PTEN polyubiquitylation for degradation or monoubiquitylation for nuclear import^{12–16}. The tight regulation of PTEN protein stability is of great importance as even a slight reduction in PTEN levels can be tremendously consequential, resulting in tumour initiation and progression, with breast cancer being the most sensitive^{17–19}.

Notably, in contrast to p53, which is degraded rapidly²⁰, PTEN seems to be a relatively stable protein. It is well documented that the

ubiquitylation of many proteins can be reversed by deubiquitylases²¹ (DUBs). There are about 90 DUBs in the human proteome consisting of five families: UCHs, USPs, OTUs, Josephins and JAMMs (ref. 21). So far, the mechanisms regulating PTEN deubiquitylation remain enigmatic. HAUSP (also known as USP7) specifically removes the monoubiquitylation of PTEN and induces its nuclear export but has minimal effect on PTEN protein stability²². The Josephin-family ataxin-3 regulates PTEN transcript abundance but not the protein stability²³. A recent study, through a screen of 30 USP members, identified USP13 as a candidate that de-polyubiquitylates and stabilizes PTEN (ref. 24). However, depletion of USP13 in cells reduced but did not abolish the PTEN ubiquitylation²⁴, suggesting that other DUB(s) might exist to function as the DUB for PTEN. It is not surprising that such an important tumour suppressor is controlled by multiple DUBs. Here, we performed a whole DUB proteome-wide screen for the regulator of PTEN and identified the OTU (ovarian tumour protease) family member OTUD3 as a potent DUB for PTEN de-polyubiquitylation. We further characterize its

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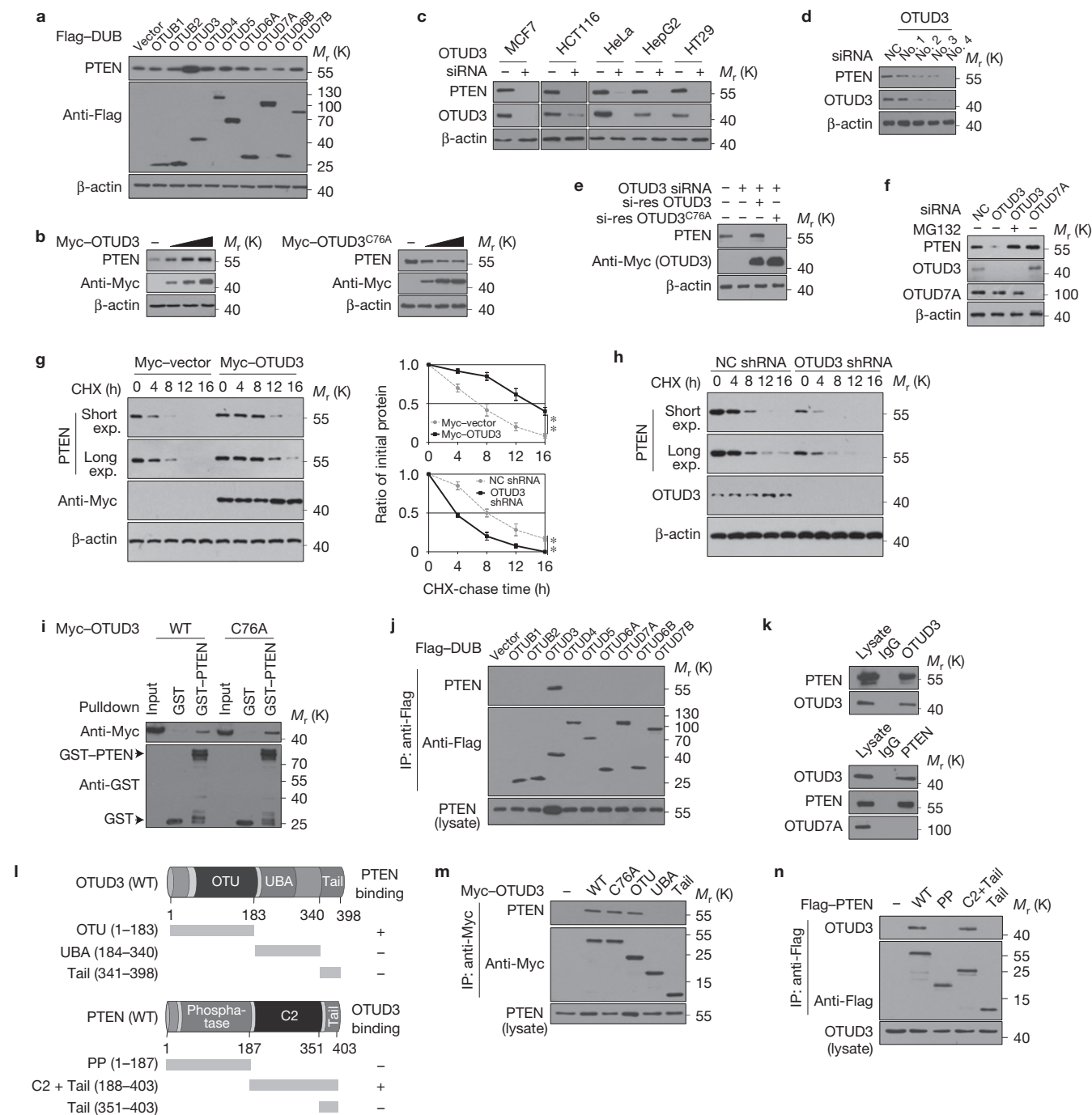


Figure 1 The deubiquitylase OTUD3 stabilizes PTEN. **(a)** The indicated OTU subfamily DUBs were each transfected into HCT116 cells. Forty-eight hours later, cell lysates were subjected to western blot. **(b)** Increasing amounts of OTUD3 WT or C76A were transfected into HCT116 cells and PTEN expression was detected. **(c)** OTUD3 was depleted by pool siRNAs in the cell lines and PTEN was analysed. **(d)** PTEN was analysed in MCF7 cells transfected with individual OTUD3 siRNA separated from pool siRNAs. **(e)** siRNA-resistant (si-res) OTUD3 WT or C76A was introduced into MCF7 cells together with OTUD3 siRNA. PTEN levels were measured. **(f)** MCF7 cells transfected with the indicated siRNA were treated with or without the proteasome inhibitor MG132 (20 μ M, 8 h), and then proteins were analysed. **(g)** MCF7 cells transfected with the indicated plasmids were treated with cycloheximide (10 μ g ml⁻¹), and collected at the indicated times for western blot. Quantification of PTEN levels relative to β -actin is shown. Results are shown as mean $\pm$ s.d. $n=3$ independent experiments. **, $P < 0.001$,

two-way ANOVA test. **(h)** Half-life analysis of PTEN in MCF7 cells stably expressing the indicated shRNA. **(i)** HCT116 cells transfected with OTUD3 WT or C76A were lysed and lysates incubated with GST or GST-PTEN-GSH-Sepharose. Proteins retained on Sepharose were blotted with the indicated antibodies. **(j)** HCT116 cells transfected with the indicated Flag-DUBs were subject to immunoprecipitation (IP) with anti-Flag antibodies. The lysates and immunoprecipitates were analysed. **(k)** MCF7 cell lysates were subject to immunoprecipitation with control IgG, anti-PTEN or OTUD3 antibodies. The immunoprecipitates were then blotted. **(l)** Overview of OTUD3 and PTEN structures. **(m,n)** HCT116 cells transfected with the indicated constructs were subject to immunoprecipitation with anti-Myc **(m)** and anti-Flag **(n)** antibodies. The lysates and immunoprecipitates were then blotted. All panels are representative results of three or more independent experiments. Statistics source data for Fig. 1g can be found in Supplementary Table 4. Unprocessed original scans of blots are shown in Supplementary Fig. 9.

pivotal role in tumour suppression by regulating PTEN. These results reveal an important missing piece in the dynamic molecular network governing PTEN stability.

RESULTS

OTUD3 maintains PTEN stability

To identify the DUB responsible for PTEN, we performed a **whole DUB proteome-wide screen** and a total of 85 DUBs were included. Each DUB was overexpressed in HCT116 cells and the expression of endogenous PTEN was analysed. This screen led us to find that OTUD3 (OTU domain-containing protein 3) remarkably upregulated PTEN levels (Fig. 1a). Other DUBs, including HAUSP, USP13 and ataxin-3, did not significantly upregulate PTEN (Supplementary Fig. 1a). **Ectopic expression** of OTUD3, but not other members of the OTU family, resulted in PTEN elevation in a **dose-dependent manner** (Fig. 1b and Supplementary Fig. 1b,c). OTUD3 regulates PTEN in a manner that is dependent on **its DUB activity** because the **catalytically inactive mutant C76A** lost its ability to upregulate PTEN, and even **exhibited a downregulating effect** (Fig. 1b, right). Depletion of OTUD3 by a **short interfering RNA (siRNA)** pool markedly decreased PTEN levels (Fig. 1c), and three independent OTUD3 siRNAs all showed significant effects (Fig. 1d). The decrease of PTEN was reversed by overexpression of an **siRNA-resistant wild-type (WT)**, but not C76A, OTUD3 or **addition of the proteasome inhibitor MG132** (Fig. 1e,f). OTUD3 depletion or overexpression had **no significant effect** on PTEN messenger RNA levels (Supplementary Fig. 1d). To prove that OTUD3 affects PTEN stability **per se**, we treated cells with the **protein synthesis inhibitor** cycloheximide. The half-life of PTEN was shortened in cells depleted of OTUD3, but prolonged in cells overexpressing OTUD3 (Fig. 1g,h). On the other hand, depletion of OTUD7A, another DUB containing OTU and UBA domains like OTUD3, showed minimal effects on PTEN (Fig. 1f and Supplementary Fig. 1e,f). These results demonstrate that OTUD3 **specifically** stabilizes PTEN in cells.

OTUD3 interacts with PTEN

Binding assays showed that OTUD3 **interacted with** PTEN *in vitro* in a manner that is **independent** of its DUB activity (Fig. 1i). Among nine OTUs we examined, **only OTUD3** was able to bind to PTEN (Fig. 1j). Importantly, endogenous OTUD3 and PTEN proteins were **co-immunoprecipitated** from lysates of five cell types (Fig. 1k and Supplementary Fig. 1g). **Deletion analysis** demonstrated that the **amino-terminal OTU region** mediated the physical interaction with PTEN (Fig. 1l,m and Supplementary Fig. 1h). Additionally, we mapped the OTUD3-binding region of PTEN to the **C2 domain** (Fig. 1l,n and Supplementary Fig. 1i).

USP13 was recently shown to stabilize PTEN through deubiquitylation²⁴. Although our screen includes USP13, we did not observe significant effects of USP13 on PTEN in our initial screen (Supplementary Fig. 1a). When the plasmid amount during **transfection** was increased, the upregulation effects of PTEN by USP13 were observed (Supplementary Fig. 2a). OTUD3 interacts with C2 of PTEN (Fig. 1n), but USP13 interacts with the **phosphatase domain** of PTEN (ref. 24). We then examined the interactions among endogenous PTEN, OTUD3 and USP13. PTEN and OTUD3 were readily co-immunoprecipitated with each other. PTEN was detected in the

anti-USP13 immunoprecipitates; **however, USP13 was not detectable in the anti-PTEN immunoprecipitates**. Also, OTUD3 and USP13 were not co-immunoprecipitated with each other (Supplementary Fig. 2b). Depletion of USP13 resulted in a downregulation of PTEN (Supplementary Fig. 2c), consistent with a previous report²⁴. We then compared the stabilizing ability of OTUD3 and USP13 on PTEN through side-by-side assays. Overexpression and depletion assays both indicated that the PTEN regulatory effects of OTUD3 were more significant than those of USP13 (Supplementary Fig. 2d,e). Ectopic expression of OTUD3 in USP13-depleted cells could upregulate PTEN, whereas overexpression of USP13 in OTUD3-depleted cells did **not** upregulate PTEN (Supplementary Fig. 2f), suggesting that OTUD3 should be a **more potent regulator** of PTEN than USP13 in our experimental condition.

OTUD3 de-polyubiquitylates PTEN

As OTUD3 is a deubiquitylase, we went on to examine the possibility that OTUD3 deubiquitylates PTEN. Indeed, *in vitro* ubiquitylation assays showed that OTUD3 directly removed the ubiquitin chain of PTEN (Fig. 2a). Ectopic expression of WT, but not C76A, OTUD3, reduced PTEN ubiquitylation in cells (Fig. 2b and Supplementary Fig. 2g). Conversely, downregulation of OTUD3, but not OTUD7A, increased PTEN ubiquitylation in cells (Fig. 2c). The enhanced PTEN ubiquitylation by OTUD3 silencing was reduced by OTUD3 overexpression (Supplementary Fig. 2h). Furthermore, OTUD3 decreased the PTEN ubiquitylation induced by the E3 ligase Nedd4-1 or WWP2 (Fig. 2d,e).

We next investigated **which type of ubiquitin chain** of PTEN was affected by OTUD3. OTUD3 effectively removed Lys 48-linked polyubiquitylation, **but not monoubiquitylation nor the non-degradative Lys 63-linked polyubiquitylation of PTEN** (Fig. 2f and Supplementary Fig. 2i). In contrast, HAUSP specifically removed monoubiquitylation of PTEN (Fig. 2f), consistent with a previous report²². As a control, overexpression of HAUSP stabilized whereas depletion of HAUSP destabilized MDM2 (Supplementary Fig. 2j–l), confirming that HAUSP used here harbours detectable DUB activity.

To extend our findings, we performed **a thorough in vitro deubiquitylation assay** of PTEN by OTUD3 and USP13 with **a series of ubiquitin mutants**. PTEN could be linked with all types of the ubiquitin in the presence of the E3 CHIP and Hsp70. Intriguingly, OTUD3 efficiently removed the Lys 6, Lys 11, Lys 27 and Lys 48 types of ubiquitin chain on PTEN whereas USP13 preferred to remove Lys 6, Lys 29 and Lys 63 chains (Fig. 2g). Neither OTUD3 nor USP13 removed the monoubiquitylation of PTEN (Lys 0 groups). Under equal amounts, OTUD3 exhibited more potency than USP13 towards WT ubiquitin chains (Fig. 2g). Taken together, OTUD3 is a specific DUB, which de-polyubiquitylates and stabilizes PTEN.

OTUD3 inhibits Akt signalling

PTEN is localized primarily in the **cytoplasm** and opposes the function of the PI(3)K/Akt pathway, but also found in the **nucleus**. We examined **which population of PTEN is regulated by OTUD3**. **Cell fractionation analysis** showed that endogenous **OTUD3** was mainly localized in the **cytoplasm** of MCF7 cells and depletion of OTUD3 resulted in **a reduction of cytoplasmic PTEN** (Fig. 3a).

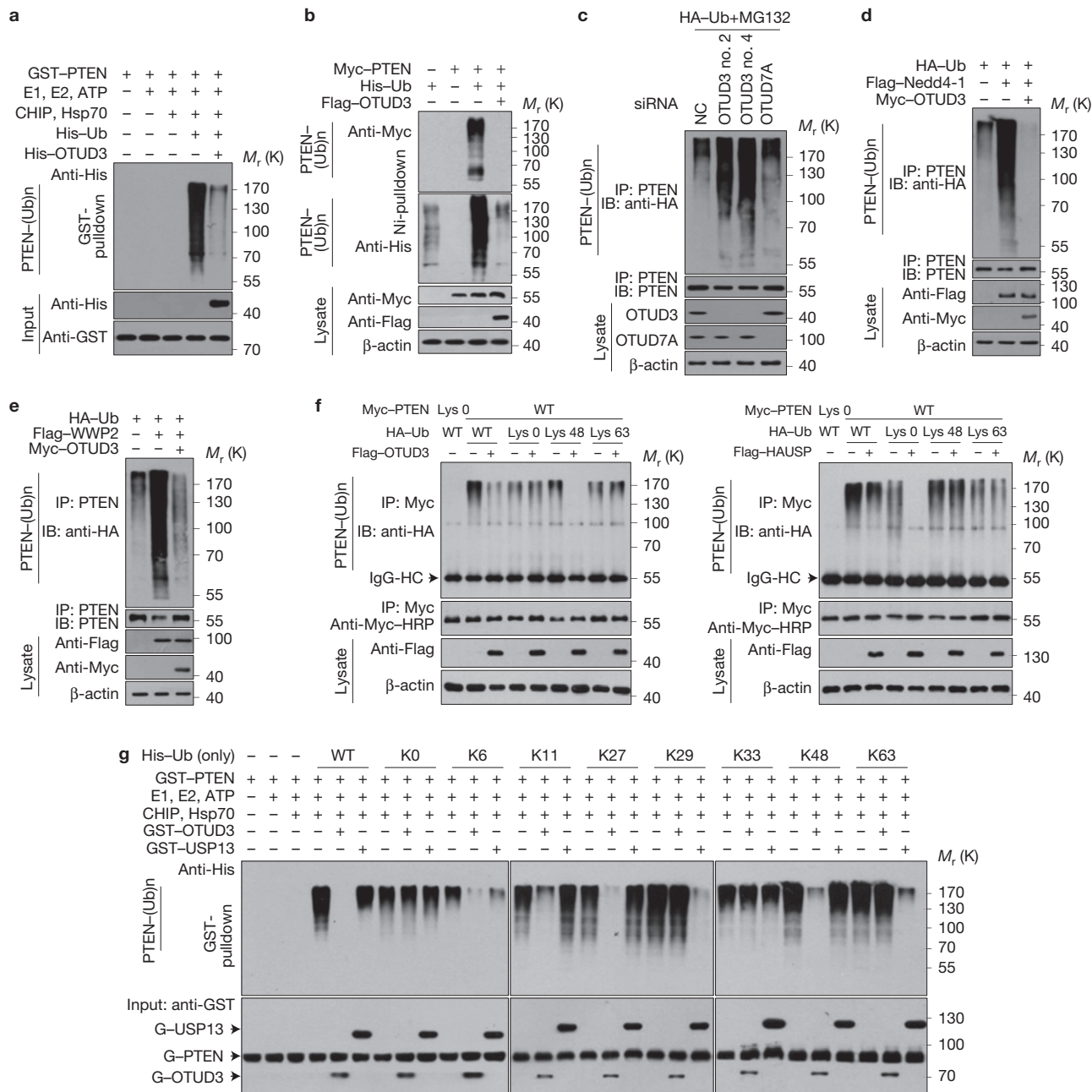


Figure 2 OTUD3 de-polyubiquitylates PTEN. **(a)** Cell-free PTEN ubiquitylation assay. The bacteria-expressed and purified GST-PTEN and His-OTUD3 proteins were incubated with commercial E1, Ube2D3 (E2), CHIP (E3), Hsp70 and ubiquitin for 2 h at 37 °C. The mixtures were subjected to GST pull-down and western blot with anti-His antibody. **(b)** MCF7 cells transfected with the indicated constructs were treated with MG132 for 8 h before collection. The whole-cell lysate was subjected to pull-down with Ni²⁺ beads and western blot with anti-Myc and anti-His antibody to detect ubiquitylated PTEN. **(c)** HCT116 cells transfected with the indicated siRNA were treated with MG132 for 8 h before collection. PTEN was immunoprecipitated with anti-PTEN and immunoblotted with anti-HA. **(d,e)** PTEN ubiquitylation was analysed in cells transfected with E3 Nedd4-1

(d) or WWP2 **(e)** together with OTUD3 or not. **(f)** The PTEN ubiquitylation linkage was analysed in HCT116 cells transfected with PTEN, OTUD3, HAUSP and the indicated ubiquitin Lys 0, Lys 48-only or Lys 63-only plasmids. PTEN Lys 0 (lysine-free mutant that cannot be ubiquitylated) was used as a control to indicate the specificity of PTEN ubiquitylation. **(g)** Cell-free PTEN ubiquitylation linkage assay. The bacteria-expressed and purified GST-PTEN, GST-OTUD3 and GST-USP13 proteins were incubated with commercial E1, Ube2D3 (E2), CHIP (E3), Hsp70 and the indicated ubiquitin WT or mutant for 2 h at 37 °C. The mixtures were subjected to GST pull-down and western blot with anti-His antibody. All panels are representative results of three or more independent experiments. Unprocessed original scans of blots are shown in Supplementary Fig. 9.

Depletion of OTUD3 or PTEN, but not OTUD7A, in PTEN WT MCF7 cells, enhanced the phosphorylation levels of Akt and the Akt downstream targets such as GSK3 β , mTOR, S6K and 4E-BP1, without

affecting their total protein levels (Fig. 3b). Reintroduction of WT, but not C76A, OTUD3 rescued the effects (Fig. 3c). Importantly, reintroduction of PTEN into OTUD3-depleted cells downregulated

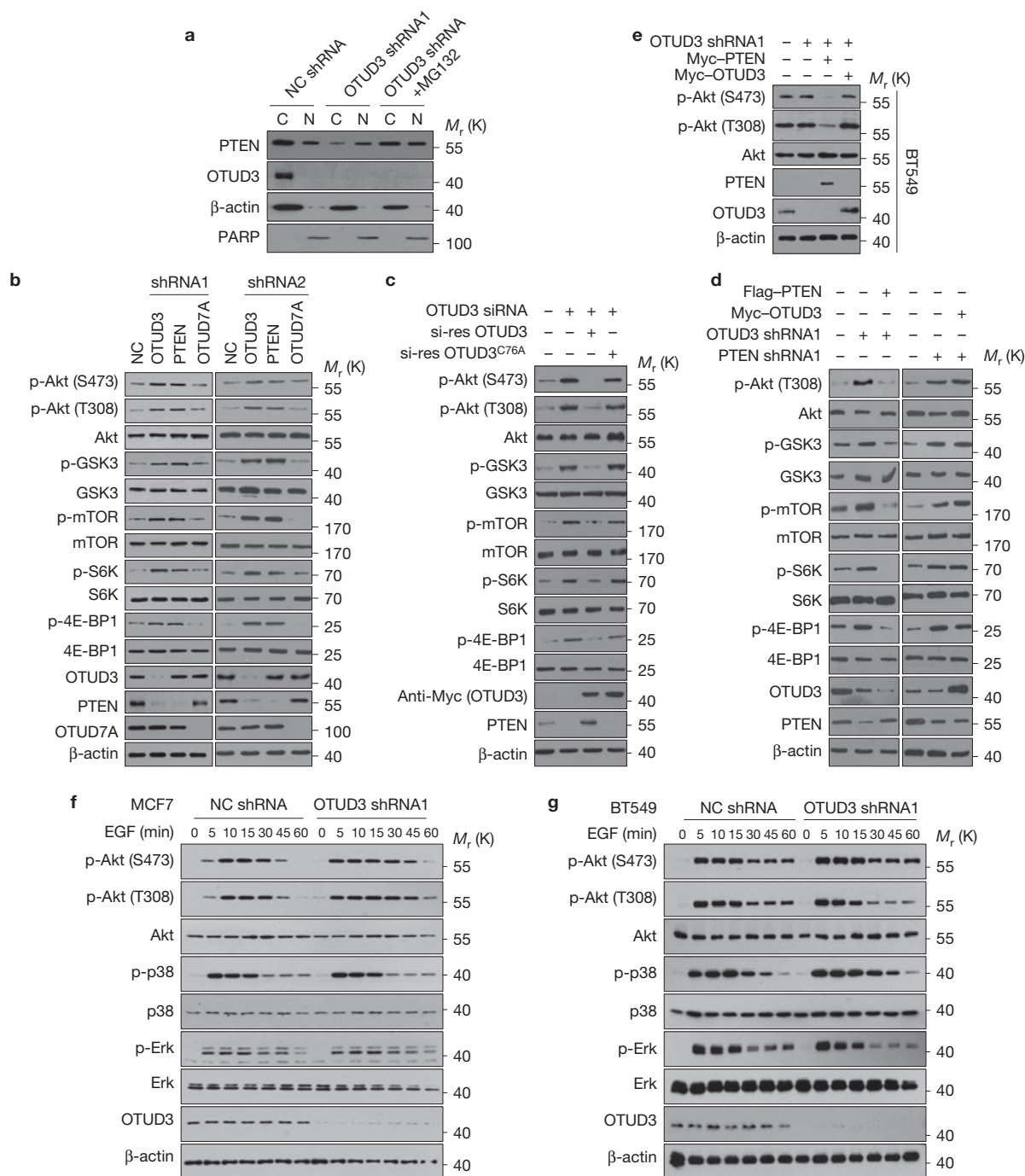


Figure 3 OTUD3 regulates cytoplasmic PTEN and inhibits Akt signalling. **(a)** MCF7 cells stably expressing the indicated shRNA were fractionated and blotted with the indicated antibodies. Actin and PARP were examined to indicate the cytoplasmic (C) and nuclear (N) extracts, respectively. **(b)** The indicated MCF7 cell lysates were subjected to analysis of the phosphorylation levels of Akt and its downstream targets. **(c)** Akt signalling was analysed in MCF7 cells transfected with OTUD3 siRNA and the siRNA-resistant (si-res) OTUD3 WT or C76A. **(d)** PTEN was introduced into the OTUD3 shRNA-transduced MCF7 cells (left);

OTUD3 was introduced into the PTEN shRNA-transduced cells (right). Akt signalling was analysed. **(e)** The PTEN or shRNA-resistant OTUD3 plasmids were transfected into PTEN-null BT549 breast cancer cells stably expressing the indicated shRNA and Akt phosphorylation was analysed. **(f,g)** OTUD3 shRNA-transduced MCF7 or BT549 cells were serum-starved and treated with EGF for various times. The cells were collected for immunoblot analysis. All panels are representative results of three or more independent experiments. Unprocessed original scans of blots are shown in Supplementary Fig. 9.

the phosphorylation levels of Akt and its targets. In contrast, OTUD3 could not rescue the effect of PTEN depletion (Fig. 3d). Additionally, OTUD3 depletion resulted in downregulation of the cell cycle

inhibitors p21Cip1, p27Kip1, p15INK4b and the pro-apoptotic Bim, as well as upregulation of the angiogenesis enhancer VEGF in MCF7 cells (Supplementary Fig. 3a,c).

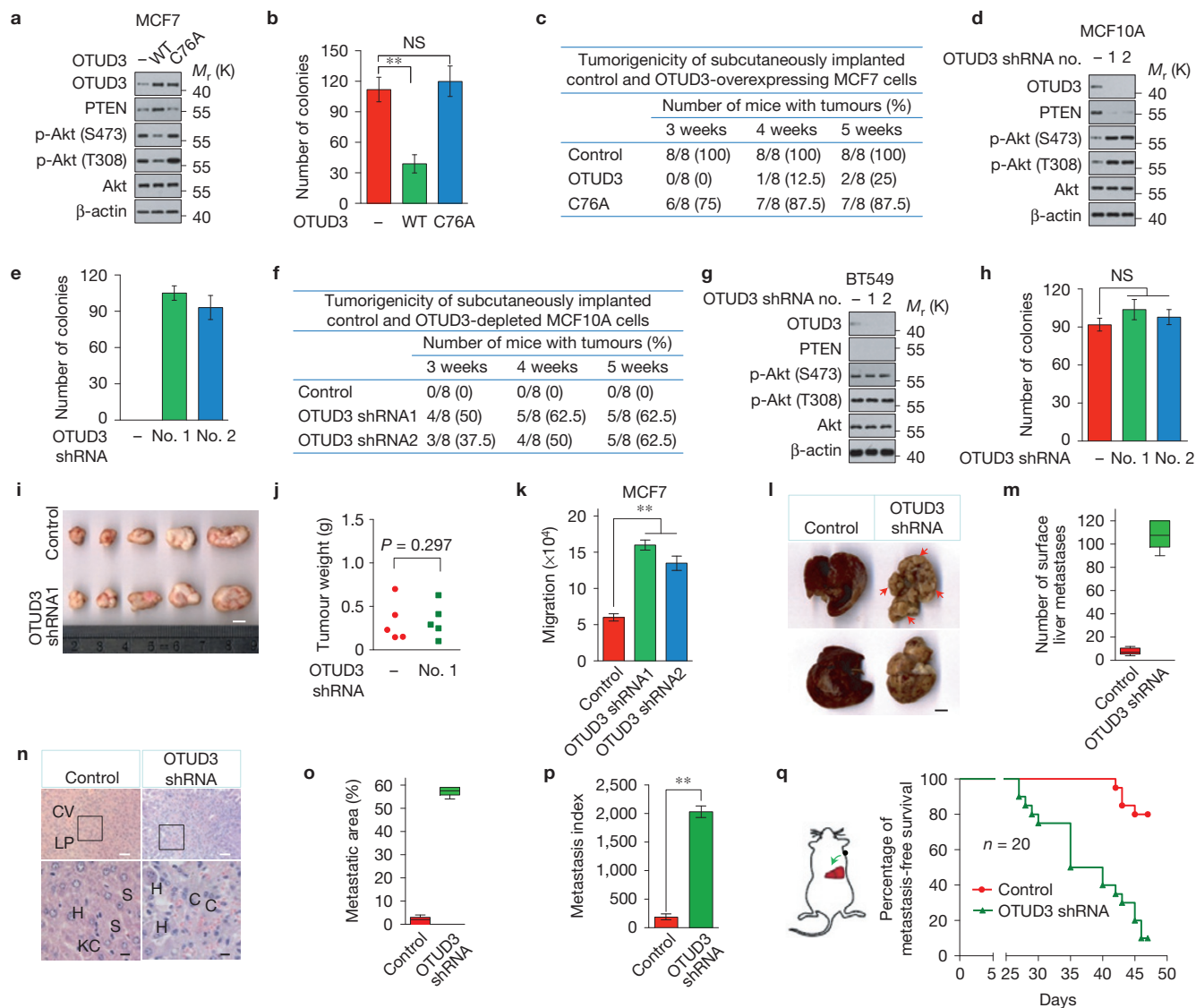


Figure 4 OTUD3 suppresses tumorigenesis. **(a)** Indicated protein levels were detected in MCF7 cells stably transfected with OTUD3 WT or C76A and MCF10A cells stably expressing the indicated shRNA. **(b)** The MCF7 cells in **a** were tested for anchorage-independent growth in a soft-agar colony assay. Viable colonies after 3 weeks were counted and data are depicted as bar graphs with mean $\pm$ s.d. $n=3$ independent experiments. **, $P<0.01$; NS, not significant; Student's t -test. **(c)** The MCF7 cells (2×10^7) were subcutaneously injected into 6-week-old female nude mice. Each cell line was injected into eight different animals, and mice forming tumours were analysed. **(d)** Indicated protein levels were detected in MCF10A cells stably transfected with the OTUD3 shRNA or control. **(e,f)** MCF10A cells in **d** were tested for colony formation on soft agar and tumour formation in nude mice (2×10^7 cells per mouse) assays. For **e** results shown are mean $\pm$ s.d. $n=3$ independent experiments. **(g)** OTUD3 shRNA-transduced PTEN-null BT549 cells were established. **(h-j)** BT549 cells in **g** were tested for colony formation **(h)** and tumour formation in nude mice (2×10^7 cells per mouse, 5 weeks) assays **(i,j)**. The tumour weights were calculated. For **h**, results shown are mean $\pm$ s.d. $n=3$ independent experiments. Data were analysed using Student's t -test **(h)** and Mann-Whitney test **(j)**. Scale bar, 5 mm. **(k)** Transwell

assay of the indicated MCF7 cells. Data are depicted as bar graphs with mean $\pm$ s.d. $n=3$ independent experiments. **, $P<0.01$, Student's t -test. **(l)** The OTUD3 shRNA or control MCF7 cells were subcutaneously implanted into one flank of athymic nude mice (2×10^7 cells per mouse, $n=10$). Representative bright-field imaging of the livers are shown and arrows indicate surface metastatic nodules. Scale bar, 5 mm. **(m)** Number of visible surface metastatic lesions in livers ($n=10$ mice). Data are depicted as bar graphs with mean $\pm$ s.d. **(n)** Liver tumours in the mice were confirmed by haematoxylin and eosin staining ($n=10$). CV, central venous; LP, liver plate; S, sinusoids; KC, Kupffer cells; H, hepatocytes; C, cancer cells. Scale bar, 100 μ m (upper panel), 20 μ m (lower panel). **(o)** Box plots showing the per cent metastatic area by section. Data were analysed using Mann-Whitney test ($P=0.0001$). **(p)** Metastatic index is also shown (**, $P<0.001$, Student's t -test). $n=10$ mice. Data are depicted as bar graphs with mean $\pm$ s.d. **(q)** Kaplan-Meier metastasis-free survival analysis. P values calculated by log-rank test ($n=20$ mice; OTUD3 shRNA, $P<0.0001$). Panels **a,d,g** are representative results of 3 or more independent experiments. Statistics source data can be found in Supplementary Table 4. Unprocessed original scans of blots are shown in Supplementary Fig. 9.

The above data suggest that OTUD3 functions upstream of PTEN. To verify whether OTUD3 regulates Akt signalling in a manner dependent on PTEN, we utilized the PTEN-null BT549 breast

cancer cells. OTUD3 depletion had no significant effect on Akt phosphorylation (Fig. 3e), or the expression of p27Kip1, p15INK4b, Bim and VEGF (Supplementary Fig. 3b,c). Furthermore, depletion of

OTUD3 led to earlier activation and prolonged duration of the Akt signalling on EGF stimulation in MCF7 cells, but not in BT549 cells (Fig. 3f,g and Supplementary Fig. 3d,e). Furthermore, the effect of OTUD3 on Akt signalling is specific because it had no effect on Erk and p38 signalling. Hence, we conclude that OTUD3 is an upstream negative regulator of the oncogenic Akt pathway.

OTUD3 suppresses tumorigenicity of human cancer cells

We propose that OTUD3, like PTEN, might function as a tumour suppressor. To test this hypothesis, we first established stable MCF7 transfectants ectopically expressing WT or C76A OTUD3 (Fig. 4a). Overexpression of WT but not C76A OTUD3 repressed tumour cell growth in soft agar and in xenografted nude mice (Fig. 4b,c). Consistently, depletion of OTUD3 in MCF7 cells promoted cell proliferation and decreased sensitivity to apoptosis (Supplementary Fig. 4a–d). Overexpression of PTEN rescued the phenotype of OTUD3 depletion but overexpression of OTUD3 did not rescue the phenotype of PTEN depletion (Supplementary Fig. 4b,d). To further assess the suppressive role of OTUD3, we sought to determine whether loss of OTUD3 would transform normal mammary epithelial cells. To this end, we established stable MCF10A transfectants and found that OTUD3 knockdown upregulated Akt phosphorylation, and induced anchorage-independent growth and tumour formation in xenografts (Fig. 4d–f), suggesting that loss of OTUD3 is sufficient to induce tumorigenicity. Importantly, OTUD3 knockdown had no significant effect on the tumorigenicity of PTEN-null BT549 cells (Fig. 4g–j), indicating the PTEN dependence for OTUD3 in governing tumorigenesis.

We next investigated the possible effect of OTUD3 on cell migration and tumour metastasis. Stable depletion of OTUD3 enhanced the migration ability (Fig. 4k) and promoted the formation of pseudopodia-like membrane protrusions of MCF7 cells (Supplementary Fig. 4e). Nude mice bearing OTUD3 shRNA tumours developed prominent metastasis to the liver, and very slight metastasis developed in the control cohort (Fig. 4l,m). Haematoxylin and eosin staining showed increased local invasion and both the area and size of liver metastases were increased in mice from the OTUD3 shRNA cohort (Fig. 4n–p and Supplementary Fig. 4f). Immunoblotting analysis confirmed that these metastatic tumours were depleted of OTUD3 and harboured increased levels of Akt phosphorylation and F-actin expression (Supplementary Fig. 4g). Metastasis-free survival analysis showed that metastasis was detected in the livers of 90% (18/20) of mice bearing OTUD3-depleted tumours, whereas only 20% (4/20) of the animals injected with control cells presented with liver nodules (Fig. 4q and Supplementary Fig. 4h). Collectively, these results demonstrate that OTUD3 is a suppressor of tumour growth and metastasis.

We further investigated the possible synergy between OTUD3 and USP13. Combined depletion of OTUD3 and USP13 resulted in a more pronounced decrease of PTEN expression and an increase of Akt phosphorylation, subsequently leading to increased cell proliferation, colonies and migration, and larger tumour formation in nude mice, compared with single depletion of either OTUD3 or USP13 (Supplementary Fig. 5a–f).

OTUD3 transgenic mice are less prone to tumorigenesis

To further investigate the stabilizing effects of OTUD3 on PTEN and the tumour suppressor role of OTUD3 *in vivo*, we generated

an OTUD3 transgenic (TG) mouse model (Fig. 5a). The TG mice exhibited reduced whole body size, and all organs we examined were smaller than those from WT mice and expressed higher levels of PTEN (Fig. 5b–d). Notably, these phenotypes are similar to those of PTEN TG mice¹⁸. The PTEN protein level in OTUD3 TG mouse embryonic fibroblasts (MEFs) was increased twofold compared with WT MEFs, although mRNAs were at similar levels (Fig. 5e). In WT MEF cells, endogenous OTUD3–PTEN interaction was easily detected (Fig. 5f). In OTUD3 TG MEFs, PTEN ubiquitylation was decreased, the half-life of PTEN was prolonged, and the duration of Akt activation was shortened (Fig. 5g–i and Supplementary Fig. 5g). Furthermore, the OTUD3 TG mice were crossed with MMTV-polyomavirus middle T antigen (PyMT) transgenic mice (Supplementary Fig. 5h), which have been widely used to study mammary tumorigenesis and metastasis^{25,26}. As expected, all of the PyMT/OTUD3 WT mice spontaneously developed breast tumours at 60–69 days of age. However, the PyMT/OTUD3 TG mice started developing tumours at 74 days and only 33.3% (2/6) of the mice developed tumours at 80 days (Fig. 5j–l). These data coherently suggest that OTUD3 is a positive regulator of PTEN *in vivo*.

Identification of OTUD3 loss-of-function mutations in human cancers

Owing to its tumour suppressive nature, we speculate that OTUD3 is mutated in human breast cancers. To test this possibility, we collected 50 paired cancer tissues and adjacent tissues of patients. The complementary DNAs were prepared and subcloned into the T vector followed by sequencing of the OTU domain of OTUD3. All the adjacent tissues of 50 cases were WT OTUD3. In the cancer tissue of three cases, both mutation and WT clones were detected, implying that one allele was mutated but the other might remain wild type. Two cases harboured an E86K mutation and the other harboured a R79T/E86K double mutation (Supplementary Fig. 6a,b). Strikingly, although all three cases carried the WT PTEN gene, the PTEN protein expression was as low as hardly detectable (Fig. 6a). We therefore tested whether OTUD3 E86K and/or R79T mutations had any effect on PTEN stability. Although both mutants could bind to PTEN, E86K lost its ability to stabilize PTEN. It, instead, destabilized PTEN and shortened its half-life (Fig. 6b–d and Supplementary Fig. 6c). The E86K mutant, like C76A, could not effectively deubiquitylate PTEN *in vivo* and *in vitro* (Fig. 6e,f).

The sequencing data suggest that the breast cancer tissue expressed both OTUD3 E86K mutant and WT protein. We speculate that E86K might compete with WT OTUD3. Chemical crosslinking assays showed that both WT and E86K OTUD3 could form homodimers and oligomers (Fig. 6g). Flag-tagged WT or E86K OTUD3 interacted with GST–OTUD3 *in vitro* (Fig. 6h). Importantly, WT OTUD3 enhanced whereas E86K inhibited the DUB activity of GST–OTUD3 towards PTEN *in vitro* and in cultured cells (Fig. 6i,j). Furthermore, when endogenous OTUD3 was depleted, the exogenous E86K had no effect on PTEN and p-Akt levels. Only when an exogenous OTUD3 was introduced, the E86K exhibited effects on PTEN and p-Akt levels (Fig. 6k), suggesting that E86K might function through interacting with and inactivating endogenous WT OTUD3 to affect PTEN stability.

In contrast to the suppression effects of WT OTUD3, ectopic E86K exhibited promotion effects on cell proliferation, colony formation,

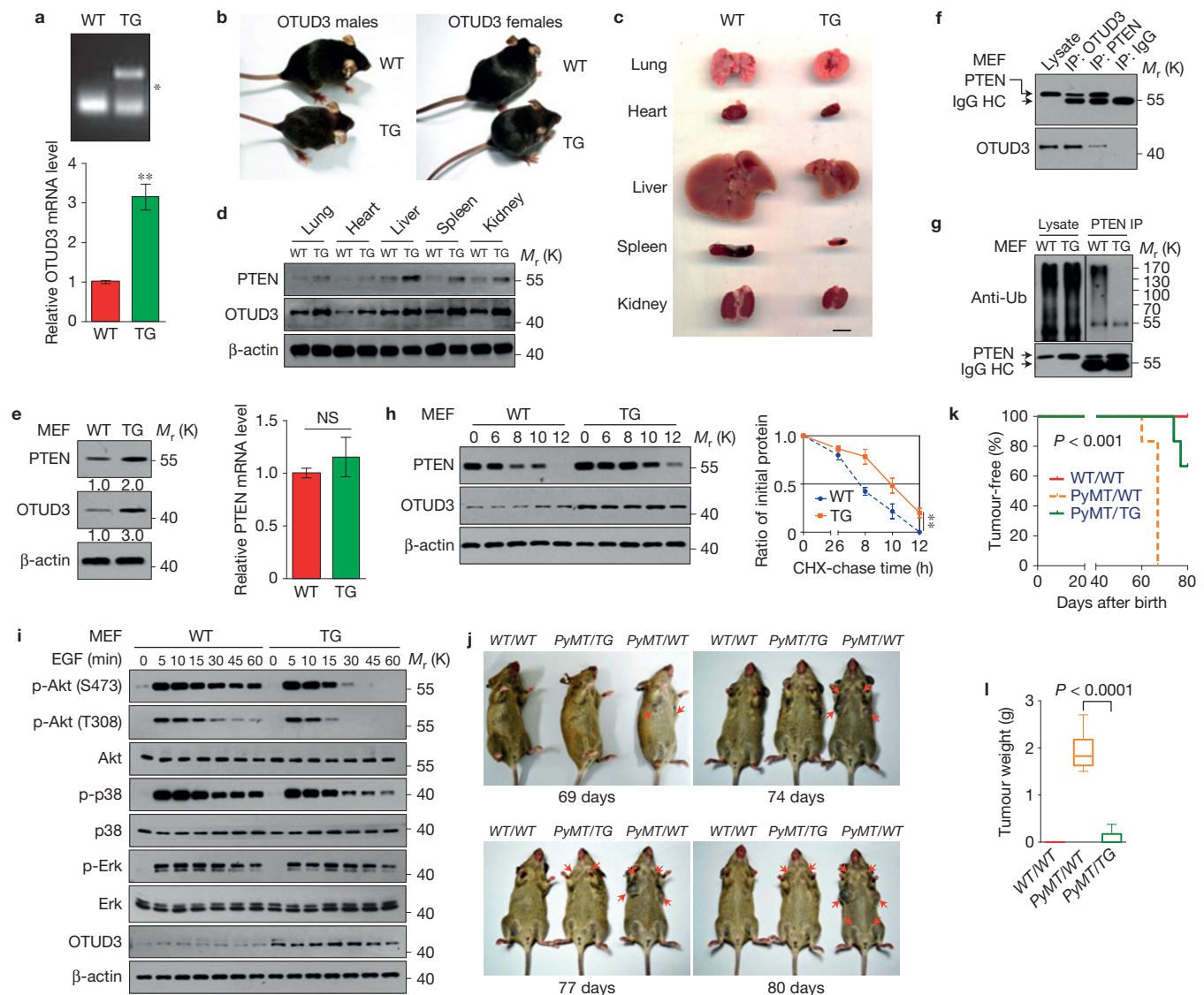


Figure 5 OTUD3 transgenic mice exhibit higher PTEN expression and are less prone to tumorigenesis. **(a)** Genotyping of the BAC-OTUD3 TG line generated. PCR analysis of genomic DNA from the embryos' tails and quantification of OTUD3 mRNA levels in MEFs obtained from WT and TG littermates **, $P < 0.01$, Student's t -test. The asterisk in the top panel indicates the TG mouse-specific band. **(b)** Effect of OTUD3 overexpression on body size. **(c)** Representative bright-field imaging of five tissues from adult WT and TG littermates. Scale bar, 5 mm. **(d)** Immunoblotting of OTUD3 and PTEN in five tissues from adult WT and TG littermates. **(e)** Analysis of PTEN protein and mRNA levels in OTUD3 WT and TG MEFs. NS, not significant; Student's t -test. **(f)** WT MEF lysates were subject to immunoprecipitation with control IgG, anti-PTEN and anti-OTUD3 antibodies to detect OTUD3-PTEN interaction. **(g)** WT and TG MEFs were treated with MG132 and the lysates were immunoprecipitated with anti-PTEN antibody followed by immunoblotting with anti-Ub. **(h)** Half-life analysis of PTEN in WT and TG MEFs. Quantification of PTEN levels relative to β -actin is shown. Results shown are mean $\pm$ s.d. **, $P < 0.001$, two-way ANOVA test. **(i)** WT

and TG MEFs were serum-starved, treated with EGF and collected for immunoblot analysis. **(j)** Representative bright-field imaging of mammary adenocarcinoma from WT/WT, PyMT/WT and PyMT/TG littermates. Data of three mice measured through 69–80 days of age are shown. **(k)** Mammary adenocarcinoma incidence in WT/WT, PyMT/WT and PyMT/TG littermates ($n = 6$ per genotype) is depicted as the percentage of tumour-free mice. Mice were considered to be tumour free until a palpable mass (>4.0 mm) persisted for longer than 4 days. P values calculated by a Gehan-Breslow-Wilcoxon test. **(l)** Total tumour weight of the mice ($n = 6$) evaluated at 80 days of age. Multiple tumours in one animal were added together. The statistical differences between groups were evaluated by a Mann-Whitney test. Panels **d–i** are representative results of 3 or more independent experiments. For **a, e**, results shown are mean $\pm$ s.d. obtained from $n = 3$ litters. Statistics source data can be found in Supplementary Table 4. Unprocessed original scans of blots are shown in Supplementary Fig. 9. For the box and whisker graphs in **l**, the box extends from the 25th to the 75th percentile, the line inside the box is the median and the whiskers are the fifth and 95th percentiles.

cell migration and formation of membrane protrusions in MCF7 cells (Fig. 6l–o). Importantly, overexpression of E86K in MCF10A cells was sufficient to induce tumour formation in nude mice (Fig. 6p). Overall, these results suggest E86K as a **dominant-negative mutant of OTUD3**.

To extend our findings, we searched the TCGA (the results shown here are partly based on data generated by the TCGA Research Network: <http://cancergenome.nih.gov>), COSMIC (<http://cancer.sanger.ac.uk/cancergenome/projects/cosmic>) and Cbioportal^{27,28} databases and found a series of OTUD3 mutations through the

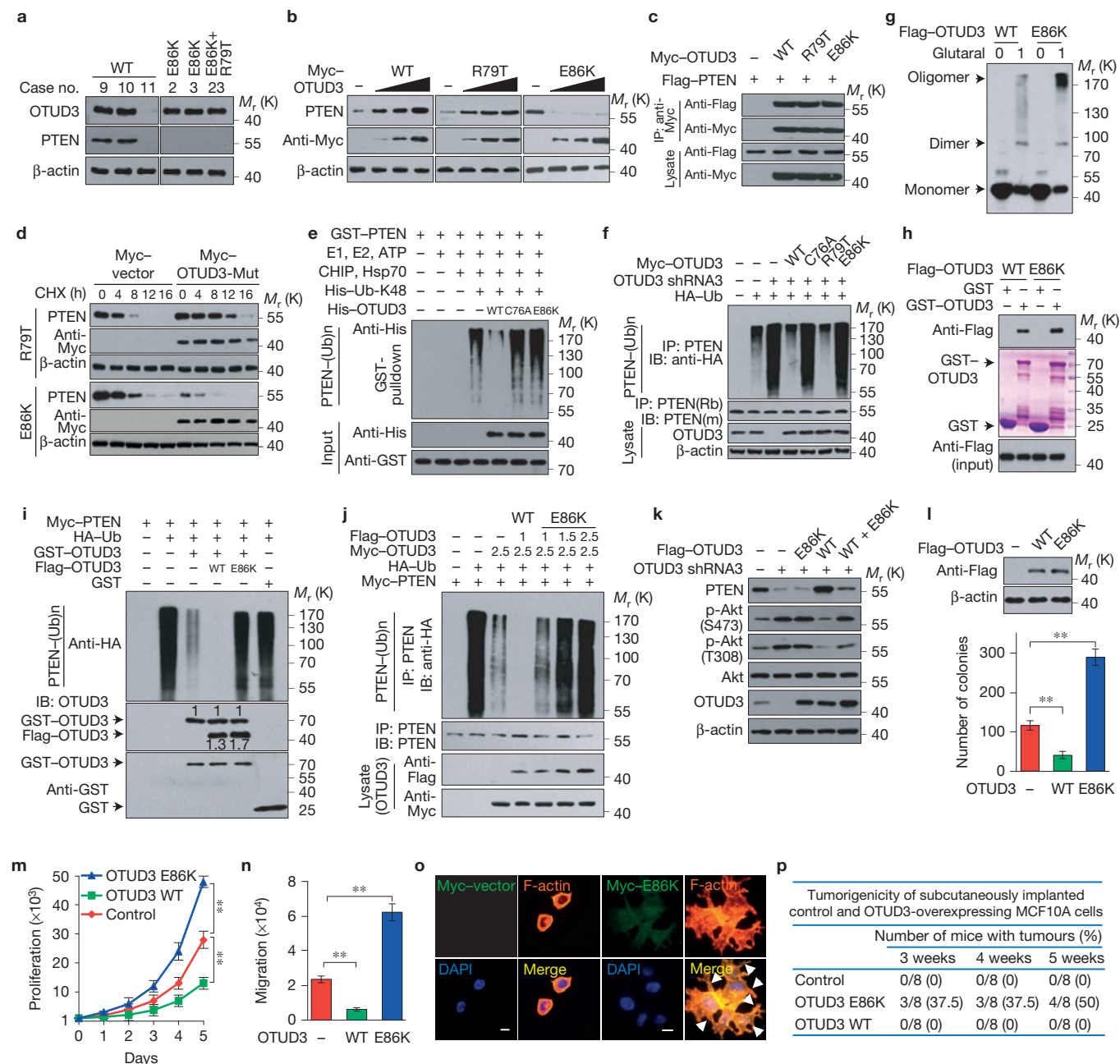


Figure 6 OTUD3 E86K mutant loses DUB function and gains oncogene function. **(a)** OTUD3 and PTEN were immunoblotted in breast cancer tissues containing WT or mutated OTUD3. **(b)** MCF7 cells were transfected with increasing amounts of OTUD3 or mutants. After 48 h, PTEN and OTUD3 were immunoblotted. **(c)** Association between the indicated OTUD3 mutants and exogenous PTEN was performed by co-immunoprecipitation assays in MCF7 cells treated with MG132 (20 μ M) for 8 h before collection. **(d)** Half-life analysis of PTEN in the presence or absence of ectopic OTUD3 mutants. **(e)** Cell-free PTEN ubiquitylation assays with OTUD3 WT, C76A and E86K. **(f)** The OTUD3-depleted MCF7 stable transfectant cells were transfected with Myc-OTUD3 WT or C76A, R79T or E86K mutant together with HA-Ub. Before collection, the cells were treated with MG132 (20 μ M) for 8 h. Cell lysates were immunoprecipitated with anti-PTEN antibody and then the immunoprecipitates were immunoblotted with anti-HA to detect PTEN ubiquitylation. **(g)** Immunoblot analysis of cell lysates derived from MCF7 that are collected in 0.5% NP40-containing PBS buffer followed by treatment with 0.01% glutaraldehyde. **(h)** Binding assays of Flag-tagged OTUD3 and mutant with GST-OTUD3 proteins. **(i)** MCF7 cells transfected

with Myc-PTEN and HA-Ub were treated with MG132 for 8 h. Ub-PTEN was immunoprecipitated with Myc-beads, and purified GST-OTUD3 from bacteria and Flag-OTUD3 WT or mutant from MCF7 cells were added to test cell-free PTEN ubiquitylation. Quantitative analysis was performed by measuring the integrated optical density. **(j)** PTEN ubiquitylation was analysed in MCF7 cells transfected with the indicated constructs. **(k)** MCF7 cells stably expressing the indicated shRNA were transfected with the indicated constructs and blotted. **(l–o)** The MCF7 cells stably expressing OTUD3 WT or mutant were measured for anchorage-independent growth in soft agar **(l)**, cell proliferation **(m)**, Transwell migration assay **(n)** and cell morphology **(o)**. **(p)** MCF10A control or OTUD3 WT or E86K-overexpressing cells were subcutaneously injected (2×10^7 cells per mouse) into 6-week-old female nude mice ($n=8$). Tumour formation was analysed. Panels **a–k, o** are representative results of 3 or more independent experiments. Data shown are mean $\pm$ s.d. $n=3$ independent experiments **(l, m, n)**. **(o)** Statistics source data can be found in Supplementary Table 4. Unprocessed original scans of blots are shown in Supplementary Fig. 9.

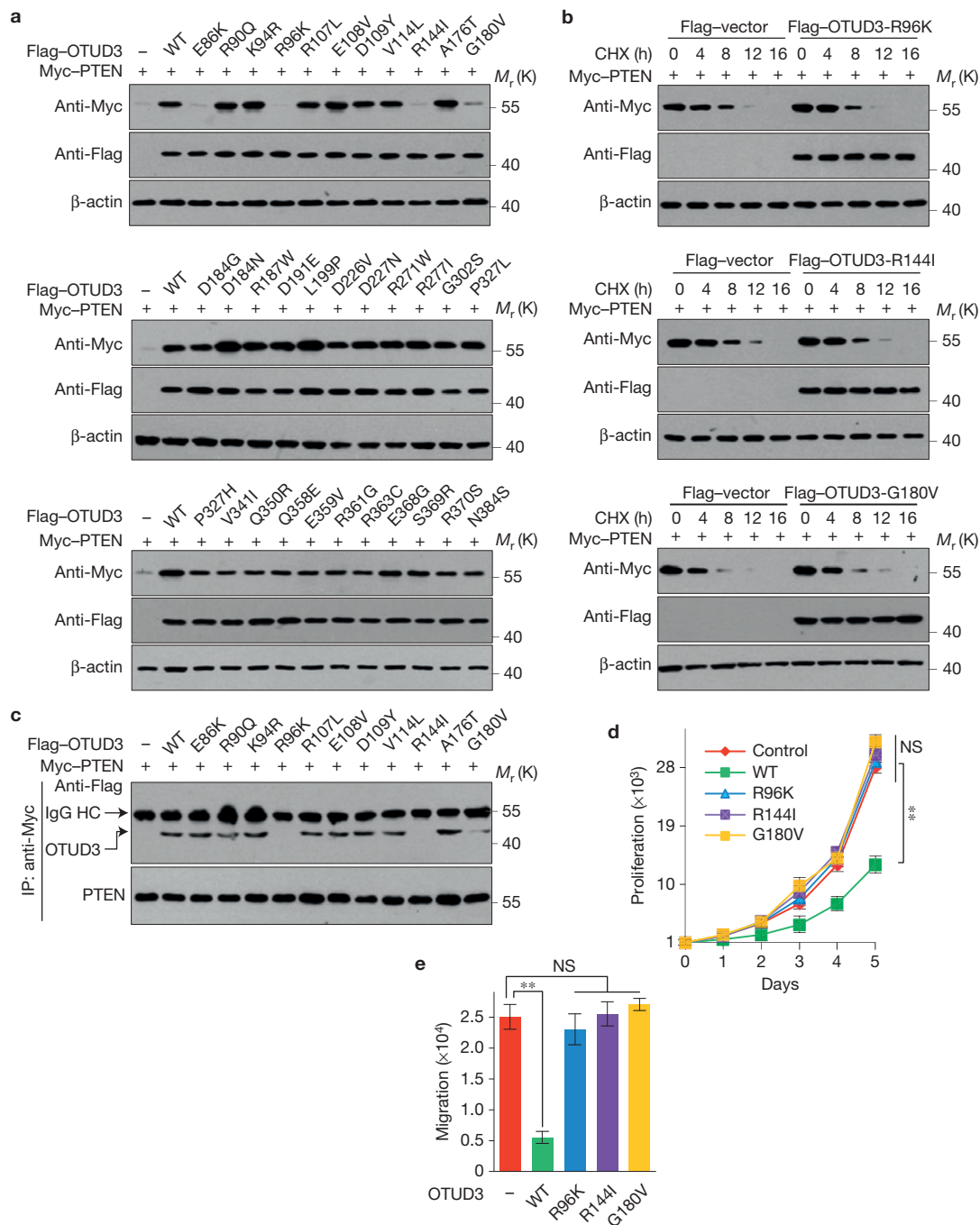


Figure 7 A subset of OTUD3 mutations in human cancer attenuate the ability to interact with and stabilize PTEN. **(a)** Mutations of OTUD3, which have been recorded in COSMIC, TCGA and Cbioportal databases, were inserted into the pFlag-CMV vector and transfected into MCF7 cells together with a small amount of Myc-PTEN. Cell lysates were then subject to western blot analysis to show PTEN protein levels. OTUD3 WT and E86K were used as the controls. **(b)** Half-life analysis of PTEN in the presence or absence of the indicated tumour-associated OTUD3 mutants R96K, R144I or G180V. **(c)** Indicated mutants of OTUD3 were overexpressed in MCF7 cells. Cell lysates were immunoprecipitated with

anti-Myc antibody. Immunoprecipitates were analysed by immunoblotting with anti-Flag and anti-PTEN antibodies. MG132 was added to the cells to maintain PTEN at similar high levels. **(d,e)** The MCF7 cells stably expressing OTUD3 WT or mutants were measured for cell proliferation **(d)**, two-way ANOVA test and Transwell migration assay **(e)**, Student's *t*-test. **, *P* < 0.01; NS, not significant. Panels **a,b** are representative results of 3 or more independent experiments. Data shown are mean ± s.d. *n* = 3 independent experiments **(d,e)**. Statistics source data can be found in Supplementary Table 4. Unprocessed original scans of blots are shown in Supplementary Fig. 9.

OTU domain to the carboxy-terminal part in various types of cancer (Supplementary Table 1). On the basis of this information, we constructed a total of 32 point-mutation plasmids, of which

10 mutations are located in the OTU domain. Intriguingly, **three mutants, that is, R96K, R144I and G180V of OTUD3**, lost the ability to stabilize the PTEN protein. In contrast, 29 other mutants acted

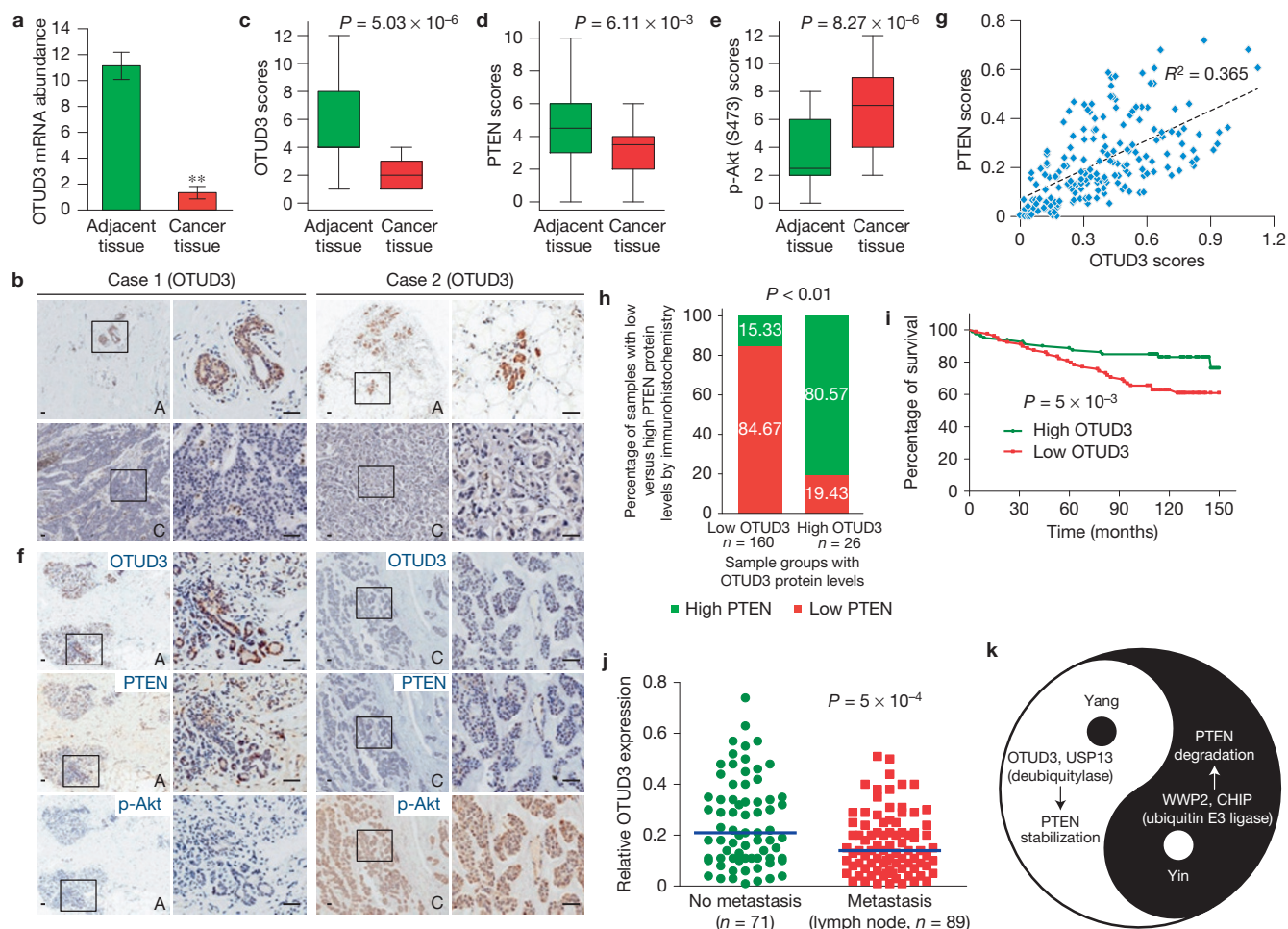


Figure 8 Positive correlation between OTUD3 and PTEN in human breast cancer. **(a)** Quantitative PCR analysis of OTUD3 mRNA in 30 breast cancer tissues with matched normal tissues. Data were analysed using two-tailed unpaired Student's *t*-test. Data are shown as mean $\pm$ s.d. of $n=3$ independent experiments. **, $P < 0.01$. **(b)** Representative images from immunohistochemical staining of OTUD3 in tumours and matched adjacent tissue from two cases in different pathological grades. The outlined areas in the left images are magnified on the right. A, adjacent tissue; C, carcinoma. Scale bars, 50 μ m. **(c–e)** OTUD3 (**c**), PTEN (**d**) and p-Akt(S473) (**e**) expression scores shown as box plots. We compared breast cancer tumours with matched adjacent normal breast epithelium using a Wilcoxon matched pairs test. $n=60$. **(f)** Representative images from immunohistochemical staining of OTUD3, PTEN and p-Akt(S473) in three serial sections of the same tumour and matched adjacent tissue. Scale bars, 50 μ m. **(g)** Regression analysis comparing OTUD3 and PTEN expression in breast cancer tissues from a second patient population. $n=186$. **(h)** Positive correlation of OTUD3 with PTEN protein levels. The 186 samples were classified into two groups (low OTUD3, 160; high OTUD3, 26) based on the OTUD3 level relative to the scores of the whole cohort. $P < 0.01$, calculated by both chi-square and

Mann–Whitney tests. **(i)** Kaplan–Meier plot of overall survival of 160 patients with breast carcinomas (two groups stratified by OTUD3 expression level medium, $n=80$). A log-rank test was used to show differences between groups. **(j)** Whisker plots show the OTUD3 expression in primary breast cancer tissues stratified by lymph node metastases. Data were analysed using Mann–Whitney test. Panels **b,f** are representative results of 60 breast cancer tissues with matched normal tissues. For **c,d,e** results shown are mean $\pm$ s.d. obtained from $n=60$ breast cancer tissues and matched normal tissues pooled from three independent experiments. For the box and whisker graphs, the box extends from the 25th to the 75th percentile, the line inside the box is the median and the whiskers are the fifth and 95th percentiles. Statistics source data can be found in Supplementary Table 4. **(k)** A 'yin–yang' control model of PTEN protein at the post-translational level. The deubiquitylases (such as OTUD3 and USP13) stabilize PTEN protein by de-polyubiquitylation (yang) whereas the ubiquitin E3 ligases (such as WWP2 and CHIP) promote PTEN degradation by polyubiquitylation (yin). These regulators orchestrate to precisely control the dynamic homeostasis of PTEN abundance. OTUD3 stabilizes PTEN to suppress tumorigenesis and metastasis. Downregulation or functional mutation of OTUD3 promotes the tumorigenesis.

like WT OTUD3 to efficiently upregulate PTEN (Fig. 7a,b). To investigate the mechanistic nature of these mutations, we tested the interaction of OTUD3 mutants with PTEN. In contrast to WT OTUD3 and other mutants, the interaction between R96K, R144I or G180V mutants with PTEN was hardly detectable (Fig. 7c), suggesting that these mutations in the OTU domain mainly impaired the binding to PTEN. Consistently, ectopic expression of OTUD3-R96K, R144I or G180V had no significant effects on

the cell proliferation and cell migration (Fig. 7d,e). Taken together, we identified a series of loss-of-function mutations of OTUD3 in human cancers.

OTUD3 expression is downregulated in human breast cancer

Last, we analysed the expression of OTUD3 in human breast cancer and the correlation between OTUD3 and PTEN. The mammary gland is among the tissues that are most sensitive to induction

of tumorigenesis by subtle PTEN reduction¹⁷. There is a 30–40% reduction in PTEN protein expression in breast carcinoma^{29,30}, whereas only a very low percentage (less than 5%) of breast cancers have been found to harbour PTEN mutations^{29,31–34}. It is therefore likely that PTEN function is compromised in breast cancer through **post-translational mechanisms**. We found that OTUD3 mRNA expression was significantly decreased in cancer tissues as compared with the matched adjacent normal tissues (Fig. 8a and Supplementary Fig. 7a). We did not observe significant correlation between OTUD3 and ER, PR or HER-2 expression in these cases (Supplementary Fig. 7b–d). Immunohistochemistry analysis of a tumour array showed that OTUD3 and PTEN were downregulated whereas the p-Akt level was upregulated in cancers compared with the adjacent tissues (Fig. 8b–e). Analysis of consecutive tissue sections showed that OTUD3 correlated with PTEN expression and inversely with p-Akt (Fig. 8f and Supplementary Fig. 7e). The correlation between OTUD3 and PTEN was further confirmed in a larger-scale array of 186 samples (Fig. 8g). Specifically, about 85% of samples with low OTUD3 exhibited low PTEN expression whereas 80% of the high OTUD3 samples exhibited high PTEN expression (Fig. 8h). Moreover, patients bearing tumours with relatively low levels of OTUD3 showed poorer overall survival than those bearing tumours with high OTUD3 (Fig. 8i). In patients with metastasis, OTUD3 levels were relatively lower than those without metastasis (Fig. 8j). Additionally, OTUD3 and USP13 exhibited a synergistic effect on PTEN expression in breast cancer samples (Supplementary Fig. 8a,b), whereas HAUSP seems to exhibit no significant correlation with PTEN (Supplementary Fig. 8c). These results suggest that OTUD3 is downregulated in breast cancers, concomitant with PTEN reduction to promote tumorigenesis and metastasis.

DISCUSSION

This study identifies OTUD3 as a specific cytoplasmic deubiquitylase for PTEN de-polyubiquitylation. Together with the previously reported E3 ubiquitin ligases and DUBs, OTUD3 contributes to the ‘yin–yang’ balance control of PTEN (Fig. 8k). HAUSP is mainly localized in the nucleus³⁵ and promotes the de-monoubiquitylation of PTEN (ref. 22) and OTUD3 and USP13 synergistically maintain PTEN stability (Supplementary Fig. 8d).

OTU DUBs regulate important cell signalling pathways. For example, OTUD5/DUBA regulates IRF3 signalling³⁶; A20 regulates NF- κ B signalling³⁷; OTUB1 regulates the DNA damage response³⁸; OTUD7B controls non-canonical NF- κ B activation in immune regulation³⁹ and OTULIN regulates linear ubiquitylation and innate immune signalling as well as angiogenesis^{40–42}. Here we show that OTUD3 specifically interacts with PTEN and regulates PI(3)K/Akt signalling. Interestingly, OTUD3 prefers to remove Lys 6, Lys 11, Lys 27 and Lys 48 linkages of PTEN ubiquitylation, the specificity of which was largely consistent with a recent study with di-ubiquitin as the substrate⁴³. Notably, OTUD3 binds to the PTEN C2 domain, a region that has been demonstrated to play a critical role in PTEN stability control^{44–46} and regulation of cell migration⁴⁷. The PTEN regulators PICT1, ROCK and Rak also bind to the C2 domain of PTEN and increase PTEN stability^{48–50}. Different from those regulators, OTUD3 possesses intrinsic DUB activity and functions through deubiquitylating PTEN.

Several lines of evidence support the concept that OTUD3 may exhibit a tumour suppressive role. First, the levels of OTUD3 mRNAs were significantly reduced in breast cancer tissues as compared with adjacent tissues, although the precise mechanisms remain to be elucidated. OTUD3 expression correlated with the PTEN protein levels in more than 200 breast cancer tissues. In addition to reduced expression, we identified a series of loss-of-function mutations of OTUD3. Second, OTUD3 functions as a suppressor of cell growth and tumour metastasis. Loss of OTUD3 was sufficient to induce tumorigenicity in mammary epithelial MCF10A cells. Depletion of OTUD3 in breast cancer cells significantly promoted cell migration and metastasis. In keeping with a tumour suppressor role of OTUD3, OTUD3 transgenic mice were less prone to tumorigenesis. Finally, OTUD3 is located at chromosome 1p36.13, which has been demonstrated to be deleted in human cancers including breast tumours. Recent studies have identified *CHD*, *KIF1B* and *PER3* as candidate tumour suppressor genes in this region^{51–53}. Future studies are needed to determine the physiological functions of OTUD3 and its synergy with other DUBs to control PTEN *in vivo*. □

METHODS

Methods and any associated references are available in the [online version of the paper](#).

Note: Supplementary Information is available in the online version of the paper

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AUTHOR CONTRIBUTIONS

The project was conceived by L.Z. The experiments were designed by L.Z., L.Y., Y.Lv, H.L., Y.Y. and F.H. Most of the experiments were performed by L.Y., Y.Lv and H.L. The establishment and phenotype analysis of transgenic mice were contributed by H.L. The protein ubiquitylation assays, the protein interaction assays and animal experiments were contributed by S.S., Y.Z., G.X., X.K. and L.W. The breast cancer sample collection and immunohistochemical analysis were contributed by H.G., Y.Li, T.Z. and D.G. The data were analysed by L.Z., L.Y., Y.Lv, H.L., Z.-X.X., Y.Y., W.W. and F.H. The manuscript was written by L.Z., L.Y. and Y.Lv.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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METHODS

Ethics statement. All animals were handled in strict accordance with the 'Guide for the Care and Use of Laboratory Animals' and the 'Principles for the Utilization and Care of Vertebrate Animals', and all animal work was approved by the Institutional Animal Care and Use Committee (IACUC) at the Beijing Institute of Radiation Medicine. The study using clinical samples including 50 paired breast cancer tumours and matched adjacent normal tissues was approved by the department of pathology at the QiLu Hospital of Shandong University. Informed consent was obtained from all subjects or their relatives.

Plasmids and antibodies. The human deubiquitylase library was purchased from OriGene. Full-length OTUD3 WT; OTU, UBA, tail truncations; and full-length C76A, R79T, E86K, R90Q, K94R, R96K, R107L, E108V, D109Y, V114L, R144I, A176T, G180V, D184G, R187W, D191E, L199P, D226V, D227N, R271W, R277I, G302S, P327L, P327H, V341I, Q350R, Q358E, E359V, R361G, R363C, E368G, S369R, R370S, N384S mutants, the siRNA-resistant OTUD3 WT, the siRNA-resistant OTUD3 C76A, full-length PTEN and deletion mutants were cloned into the pCMV-Myc and pFlag-CMV-2 vectors as indicated. GST-tagged PTEN and GST-tagged OTUD3 were cloned into the pGEX-4T-2 vector. A series of S-protein/Flag/SBP triple-tagged (SFB)-PTEN and PTEN-truncated plasmids were gifts from S. Maddika. Antibodies used in immunoblotting: anti-PTEN (Abcam, ab32199, 1:184, 1:500 dilution), anti-OTUD3 (Abcam, ab107646, 1:100) and anti-OTUD7A (Abcam, ab82490, 1:100) were purchased from Abcam., Anti-MDM2 (Santa Cruz, sc-965, SMP14, 1:100), anti-GST (Santa Cruz, sc-374171, A-6, 1:1,000), anti-actin (Santa Cruz, sc-1616, 1:1,000), anti-p-Erk (Santa Cruz, sc-136521, 1:1,000) and normal IgG (Santa Cruz, sc-2025) control were from Santa Cruz Biotechnologies. Anti-OTUD3 (Sigma, HPA028543, 1:100 for immunohistochemistry) and anti-Flag (Sigma, F7425, 1:1,000) were from Sigma. Anti-Myc (MBL, M047-3, 1:1,000) and anti-HA (MBL, M180-3, 1:1,000) were from MBL. Anti-Akt (CST, no. 9272, 1:1,000), anti-pThr308-Akt (CST, no. 9275, 1:1,000), anti-pSer473-Akt (CST, no. 4060, D9E, 1:1,000), anti-GSK3 (CST, no. 9325, 1:1,000), anti-pSer9-GSK3 (CST, no. 9336, 1:1,000), anti-mTOR (CST, no. 2983, 7C10, 1:1,000), anti-pSer2448-mTOR (CST, no. 2971, 1:1,000), anti-p70 S6K (CST, no. 9202, 1:1,000), anti-pThr389-p70 S6K (CST, no. 9209, 1:1,000), anti-4E-BP1 (CST, no. 9452, 1:1,000), anti-pSer65-4E-BP1 (CST, no. 9451, 1:1,000), anti-p38 (CST, no. 8690, D13E1, 1:1,000), anti-p-p38 (CST, no. 4511, D3F9, 1:1,000), anti-Erk and anti-USP13 (CST, no. 12577, D4P3M, 1:1,000) antibodies were purchased from Cell Signaling Technology. Anti-PTEN (CST, no. 9188, 1:100 for immunohistochemistry) was from Cell Signaling Technology. To avoid interference of the IgG heavy chain with the PTEN band, Clean-Blot IP Detection Reagents and Kit (catalogue no. 21230, purchased from Thermo Scientific) was used.

RNA interference. Smart pool siRNAs against OTUD3 (siRNA 1, 5'-GCGCUGA ACUGUACUAGUA-3'; siRNA 2, 5'-GGCCAGCCCUAGUGAAGAA-3'; siRNA 3, 5'-GAAAUACAGGGCUUAAAUGA-3'; siRNA 4, 5'-TCGCAAAGGTCAACA ACA-3') were purchased from Dharmacon. For HAUSP reduction, the siRNAs (siRNA 1, 5'-AAUGUGGCCUGAGUGAUGGA-3'; siRNA 2, 5'-AAGUCUUU GUACAGGCGGAUG-3'; siRNA 3, 5'-UGACGUGUCUCUUGAUAAAUU-3') were used. Control siRNA and the on-target individual siRNAs against OTUD3 were synthesized by Shanghai GenePharm. The USP13 shRNA clone was from Open Biosystems through MD Anderson's shRNA and ORFeome Core, V3LHS_356746 (5'-CCATGTGAACAATGATCCA-3'). Transfection was carried out according to the manufacturer's protocol. After 48 h, cells were washed with phosphate-buffered saline, lysed directly into TNE lysis buffer (50 mM Tris-HCl pH 7.5, 150 mM NaCl, 1 mM EDTA and 1% Nonidet P40) and resolved by 10% SDS-polyacrylamide gel electrophoresis.

Lentivirus infection. Lentiviruses carrying shRNA targeting human OTUD3, PTEN or OTUD7A lentiviral vectors (GV112) were from GeneChem. Lentiviruses carrying overexpressing lentiviral vectors were constructed by ourselves. The viruses were used to infect cells in the presence of Polybrene. Forty-eight hours later, MCF7 or MCF10A cells were cultured in medium containing puromycin for the selection of stable clones. The clones stably knocking down OTUD3, PTEN or OTUD7A were identified and verified by western blotting. The shRNA sequences are below. OTUD3 no. 1: 5'-TGGAAATCAGGGCTTAAAT-3'; no. 2, 5'-GAGTTACATCGCATAT C-3'; no. 3, 5'-CGTCTGCCATCGCATATTA-3'; PTEN no. 1, 5'-GCGCTATGTG TATTATTAT-3'; no. 2, 5'-TGAGATATAAGACAAGGA-3'; OTUD7A no. 1, 5'-G CGAGAACTGTGCGTTCTA-3'; no. 2, 5'-AGCAATTCTAACAGCAATA-3'; the non-targeting control, 5'-TTCTCCGAACGTGTACGT-3'.

Cell transfections, immunoprecipitation and immunoblotting. MCF7, HCT116, HEK293T, HeLa and BT549 cells were purchased from ATCC, and authenticated by STR profiling and tested for mycoplasma contamination by GENEWIZ. Cells were transfected with various plasmids using TurboFect

in vitro transfection reagent (Fermentas) or Lipofectamine 2000 (Invitrogen) reagent according to the manufacturer's protocol. For immunoprecipitation assays, cells were lysed with HEPES lysis buffer (20 mM HEPES, pH 7.2, 50 mM NaCl, 0.5% Triton X-100, 1 mM NaF and 1 mM dithiothreitol) supplemented with protease-inhibitor cocktail (Roche). Immunoprecipitations were performed using the indicated primary antibody and protein A/G agarose beads (Santa Cruz) at 4 °C. The immunocomplexes were then washed with HEPES lysis buffer four times. Both lysates and immunoprecipitates were examined using the indicated primary antibodies followed by detection with the related secondary antibody and the SuperSignal west pico chemiluminescence substrate (Thermo).

GST pulldown assays. Bacterial-expressed GST, GST-OTUD3 or GST-PTEN bound to glutathione-Sepharose 4B beads (from GE) was incubated with Myc-OTUD3, SFB-PTEN or their truncates expressed in HEK293T cells for 2 h at 4 °C. Then the beads were washed with GST binding buffer (100 mM NaCl, 10 mM Tris, 50 mM NaF, 2 mM EDTA, 0.5 mM Na₃VO₄ and 1% Nonidet P40) four times and proteins were eluted, followed by western blotting.

Protein half-life assay. For the PTEN half-life assay, Lipofectamine 2000 transfection was performed when MCF7 cells in 2 cm plates reached about 60% confluence. Plasmids encoding OTUD3 WT and mutants (R79T, E86K), PTEN or siRNAs were used in transfection as indicated in individual experiments. Twenty-four hours later, the cells were treated with the protein synthesis inhibitor cycloheximide (Sigma, 10 µg ml⁻¹) for the indicated durations before collection. The cells stably expressing the indicated shRNAs only were treated with the protein synthesis inhibitor cycloheximide (Sigma, 10 µg ml⁻¹) for the indicated durations before collection.

Cell fractionation. HCT116 cells stably expressing the indicated shRNAs were treated for 10 h with or without the proteasome inhibitor MG132 (20 µM) before being collected. Cytoplasmic and nuclear fractions were separated by using NE-PER Nuclear and Cytoplasmic Extraction Reagents Kit (Thermo).

In vitro ubiquitylation of PTEN. The reaction was carried out at 37 °C for 2 h in 20 µl reaction buffer (20 mM Tris-HCl, pH 7.2, 5 mM MgCl₂, 50 mM NaCl, 1 mM 2-mercaptoethanol, 10% glycerol) containing the following components: 50 µM of ubiquitin WT or Lys 48 only, 100 nM human E1, 2 µM Ube2D3, 2 µM CHIP, 2 µM Hsp70, 2 mM ATP (all from Ubiquitin-Proteasome Biotechnologies, Cat. no. J5110), 5 µM of GST-PTEN and His-OTUD3. The reaction was terminated by adding 0.4 ml pulldown buffer (20 mM Tris-HCl, pH 7.5, 500 mM NaCl, 1% Triton X-100, 0.02% BSA, and 5 mM β-mercaptoethanol). After addition of 40 µl of glutathione-Sepharose, the samples were rotated at 4 °C for 6 h. The beads were washed with 1 ml of pulldown buffer three times. The proteins bound to beads were released by boiling in 50 µl of 2× SDS-PAGE sample buffer for 10 min. The samples were then resolved by 8% SDS-PAGE followed by immunoblot analysis using a monoclonal anti-His antibody.

In vivo PTEN ubiquitylation assay. For *in vivo* PTEN ubiquitylation assay, OTUD3, NEDD4-1, WWP2, HAUSP and HA-ubiquitin were transfected into MCF7 cells with Lipofectamine 2000. Twenty-four hours later, the cells were treated with 20 µM of the proteasome inhibitor MG132 (Calbiochem) for 8 h. The cells were washed with PBS, pelleted, and lysed in HEPES buffer (20 mM HEPES, pH 7.2, 50 mM NaCl, 1 mM NaF, 0.5% Triton X-100) plus 0.1% SDS, 20 µM MG132 and protease-inhibitor cocktail. The lysates were centrifuged to obtain cytosolic proteins and incubated with anti-PTEN antibody (Abcam) for 3 h and protein A/G agarose beads (Santa Cruz) for a further 8 h at 4 °C. Then the beads were washed three times with HEPES buffer. The proteins were released from the beads by boiling in SDS-PAGE sample buffer and analysed by immunoblotting with anti-HA monoclonal antibody (MBL).

Fluorescence microscopy. For detection of subcellular localization by immunofluorescence, after fixation with 4% paraformaldehyde and permeabilization in 0.2% Triton X-100 (PBS), cells were incubated with the indicated PTEN and OTUD3 antibodies (dilution 1:50; Abcam) for 8 h at 4 °C, followed by incubation with TRITC-conjugated or FITC-conjugated secondary antibody (dilution 1:200; Cwbio) for 1 h at 25 °C. F-actin was detected with Acti-stain 555 phalloidin (Cytoskeleton). The nuclei were stained with DAPI (Sigma), and images were visualized with a Zeiss LSM 510 Meta inverted confocal microscope.

Apoptosis assays. Cells stably knocked down for OTUD3 were transfected with Myc-PTEN; meanwhile, cells stably knocked down for PTEN were transfected with Myc-OTUD3. Forty-eight hours later, transfected cells were treated with cisplatin (10 µM, 24 h, Sigma) or dimethylsulphoxide. The apoptotic cells were then washed

with PBS and stained with fluorescein isothiocyanate–Annexin V and propidium iodide according to the manufacturer's protocol (Beijing Biosea biotechnology Annexin V Kit). Apoptotic cells (Annexin V positive, propidium iodide negative) were then determined by flow cytometry.

In vitro proliferation assay. Cells were assayed for MTS (3-(4,5-dimethylthiazol-2-yl)-5-(3-carboxymethoxyphenyl)-2-(4-sulphophenyl)-2H-tetrazolium, inner salt Sigma) reduction. Cells were plated in 96-well plates (100 µl cell suspensions, 1×10^4 cells ml⁻¹). Twenty-four hours later, 0.05 mg ml⁻¹ MTS reagent (Promega) was added to each well and incubated at 37 °C for 4 h followed by absorbance measurement at 490 nm. The values were standardized to wells containing media alone.

Invasion assays. The assay was performed in an invasion chamber consisting of a 24-well tissue culture plate with 12 cell culture inserts (Corning). A cell suspension in serum-free culture medium (10⁶ cells ml⁻¹) was added to the inserts, and each insert was placed in the lower chamber containing 10% FBS (Hyclone) culture medium. After 48 h, MTS was added into the wells and incubated with the cell inserts for 4 h. The attenuation was measured spectrophotometrically at 490 nm.

Soft-agar colony formation assay. Cells were resuspended in DMEM containing 0.35% low-melting agarose (Sigma) and 10% FBS and seeded onto a coating of 0.7% low-melting agarose in DMEM containing 10% FBS. Plates were incubated at 37 °C and 5% CO₂ and colonies were scored 3 weeks after preparation. Colonies larger than 0.1 mm diameter were scored as positive.

Tumour growth and metastasis assay. BALB/c nude mice (6 weeks old, 18.0 ± 2.0 g) were randomly divided into indicated groups; the mice in the groups were subcutaneously injected with the indicated cells stably expressing the indicated shRNAs or constructs (2×10^6 cells in a volume of 100 µl PBS), or injected through lateral tail vein. Tumour growth was monitored twice a week by Vernier calliper measurement of two perpendicular tumour diameters. When the mouse was about to die, it was dissected to observe the metastatic organs. Haematoxylin and eosin staining was reviewed to identify the cancer tissue and normal tissue.

Immunohistochemical staining of metastatic livers. The liver tissues were washed with PBS and fixed overnight with 4% paraformaldehyde/PBS at 4 °C. The samples were dehydrated, embedded in paraffin, and sectioned into 3.5-µm-thick transverse sections. The sections were deparaffinized, treated with 3% H₂O₂ for 15 min, then microwaved in 10 mM citric sodium (pH 6.0) for 15 min to unmask antigens, rinsed in PBS, and then incubated with rabbit polyclonal anti-p-Akt (Ser473; Cell Signaling Technology, 1:50), anti-F-actin (Bioss, 1:100), anti-OTUD3 (Abcam, 1:50), anti-PTEN (ABclonal Biotechnology, 1:100) and anti-Akt (Santa Cruz Biotechnologies, 1:100) overnight at 4 °C. Secondary antibody was incubated for 30 min at 37 °C. Signal amplification and detection was performed using the DAB system according to the manufacturer's instructions.

BAC transgenesis. The *OTUD3* transgenic mice were generated by Cyagen Biosciences. For transgenesis, a large genomic insert (152.84 kb) containing the entire *Otud3* locus and cloned into the BAC vector pBACe3.6 was isolated from a mouse BAC genomic library (BAC RP23-289114 clone; RPCI library, C57BL/6J). After digestion with EcoRI, linearized BAC DNA was used for microinjection into the pronuclei of fertilized oocytes, derived from intercrosses between (C57BL/6 × CBA) F1 mice. Transgenic mice were identified by PCR of tail-tip genomic DNA. The sense and antisense primers were 5'-GGCCGCTAATACGACTCACT-3' and 5'-TTCTGTGGCTGAAAACCCCT-3', respectively. Amplification of the transgene results in a 528 bp PCR product. Briefly, 2 µl of genomic DNA was amplified in 10 × PCR buffer (TaKaRa), 200 µM dNTPs (Sigma) and 0.3 µM of each primer in a total volume of 20 µl with 1 U of Taq polymerase (TaKaRa). After an initial denaturation at 94 °C for 3 min, samples were amplified for 35 cycles, consisting of denaturing at 95 °C for 30 s, annealing at 57 °C for 30 s, followed by extension at 72 °C for 45 s. After the last cycle, samples were incubated for 5 min at 72 °C and resolved in 1.5% TAE-agarose gels. A founder mouse was mated with a wild-type C57BL/6 mouse to obtain the F1 generation. To obtain the PyMT/OTUD3 TG mice, MMTV–PyMT mice (PyMT/WT mice, FVB/N background) were crossed with OTUD3/WT mice. The mating strategy is illustrated in Supplementary Fig. 5h; PCR was used to screen PyMT/OTUD3 TG mice. The sense and antisense primers for MMTV–PyMT were 5'-GGAAGCAAGTACTTCACAAGGG-3', 5'-GGAAGTCACTAG GAGCAGGG-3', 5'-CAAATGTTGCTGTCTGGTG-3' and 5'-GTCAGTCGAGT GCACAGTTT-3' respectively. The PCR amplifications of the MMTV–PyMT mice are 200 bp and 556 bp, whereas the PCR product of wild-type mice is only 200 bp. For spontaneous mammary tumour experiment, female WT/WT ($n=6$), PyMT/TG ($n=6$) and PyMT/WT ($n=6$) mice were used to observe breast carcinogenesis from 6 weeks old. On the 80th day after birth, all of the female PyMT/TG and PyMT/WT

mice were subjected to euthanasia and the spontaneous mammary tumours were taken out for experimental comparison.

Assays with MEFs. MEFs were isolated from embryonic day (E) 12.5 embryos and cultured in DMEM medium supplemented with 10% FBS, penicillin/streptomycin, and 2 mM L-glutamine using standard techniques¹⁸.

Cohort and immunohistochemistry. Fifty paired breast cancer tumours and matched adjacent normal tissues were approved by the department of pathology at the Qilu Hospital of Shandong University. The breast tissue microarrays, purchased from Shanghai Outdo Biotech Company and Xi'an Alenabio Company, contain 70 pairs of breast cancer tumours together with matched adjacent normal breast epithelium, 216 of invasive ductal carcinoma (IDC), 170 of breast tumours with follow-up (range 2 to 150 months) and 70 of breast cancer tumour samples, respectively. Immunohistochemistry was performed by using the avidin–biotin complex method, including heat-induced antigen–retrieval procedures. Incubation with antibodies against PTEN (1:150 dilution; Cell Signaling, D4.3), OTUD3 (1:100 dilution; Abcam), phospho-Akt (Ser 473; 1:50 dilution; Cell Signaling, D9E), USP13 (1:100 dilution; Cell Signaling, D4P3M) and HAUSP (1:50 dilution; Cell Signaling, D17C6) was carried out at 4 °C for 18 h. All staining was assessed by a quantitative imaging method (inForm, PerkinElmer) that makes continuous measures and pathologists blinded to the origin of the samples and subject outcome synchronously. The widely accepted German semiquantitative scoring system in considering the staining intensity and area extent was used. Each specimen was assigned a score according to the intensity of the nucleic, cytoplasmic, and/or membrane staining (no staining = 0; weak staining = 1, moderate staining = 2, strong staining = 3) and the extent of stained cells (0% = 0, 1–24% = 1, 25–49% = 2, 50–74% = 3, 75–100% = 4). The final immunoreactive score was determined by multiplying the intensity score with the extent of score of stained cells, ranging from 0 (the minimum) to 12 (the maximum).

Screening the mutation in the breast cancer. Total breast cancer tissue RNA was prepared using the SV Total RNA Isolation System (Promega) following the manufacturer's instructions. One microgram of total RNA was subjected to reverse transcription to synthesize cDNA using the Transcriptor First Strand cDNA Synthesis Kit (Roche). A 20 µl volume reaction consisted of 1 µl reverse transcription product and 500 nM of each primer. Total genomic DNA was prepared; a 20 µl volume reaction consisted of 0.1 µg total genomic DNA and 500 nM of each primer. Q5 High-Fidelity DNA Polymerase was used for mutation selection and the primers are described in Supplementary Table 2.

Immunohistochemical evaluation of outcome tissue microarrays (TMAs). Multispectral bright-field images were acquired from each TMA core using an automated MSI system (Vectra, PerkinElmer) and then each was unmixed into their two individual component images. Pathologist-trained morphologic software (inForm, PerkinElmer) was applied to segment each image into tumour, stromal and blank regions to automatically divide each tissue category. This training was done by drawing representative training regions on 20% of the images in the cohort and allowing the software to 'learn' to recognize the morphologies of interest. This is an iterative process (draw regions; train; view resulting segmentation; redraw regions; and so on) that is done until the pathologist is satisfied with the results on a blind test set (images that were not included in the training set). The resulting image segmentation algorithm is then applied to each image. Once the images were segmented, a nuclear segmentation was performed using the unmixed haematoxylin image to find each nucleus. Each image's nuclei were divided up into those within the tumour region and those within the stromal region. The final score was given based on the intensity of the DAB channel on each cell within the tumour area of each TMA core. All samples were also reviewed by pathologists to ensure that assigned scores were appropriate.

RT-PCR and quantitative PCR. Total cell RNA was prepared using TriZol reagent (Invitrogen) following the manufacturer's instructions. Five micrograms of total RNA was subjected to reverse transcription to synthesize cDNA using the First Strand cDNA Synthesis Kit (TOYOBO) or RevertAid First Strand cDNA Synthesis Kit (Thermo). A 20 µl volume reaction consisted of 1 µl reverse transcription product and 250 nM of each primer. The primers used for the indicated gene products are described in Supplementary Table 3.

Statistical analysis. All results are shown as the mean ± s.d. of multiple independent experiments, not technical replicates. Detailed n values for each panel in the figures are stated in the corresponding legends. Statistic source data are listed in Supplementary Table 4. A Student's t -test, a Mann–Whitney test (for two group comparisons) or a Kruskal–Wallis one-way ANOVA followed by Dunn's multiple comparison tests (for more than two group comparisons) was

used for statistical analyses. Survival curves were derived from Kaplan–Meier estimates, and we compared the curves using log-rank tests. All statistical analyses were performed with GraphPad Prism 5 and SPSS 19.0 software. All statistical tests were two-sided, and *P* values <0.05 were considered to be statistically significant.

No statistical method was used to predetermine sample size. In the nude mouse experiments, the mice were randomly assigned. The investigators were blinded to experiments and outcome assessment during nude mouse and TG mouse model experiments. Other experiments were not randomized, and the investigators were not blinded to allocation during experiments and outcome assessment.

Correction notice

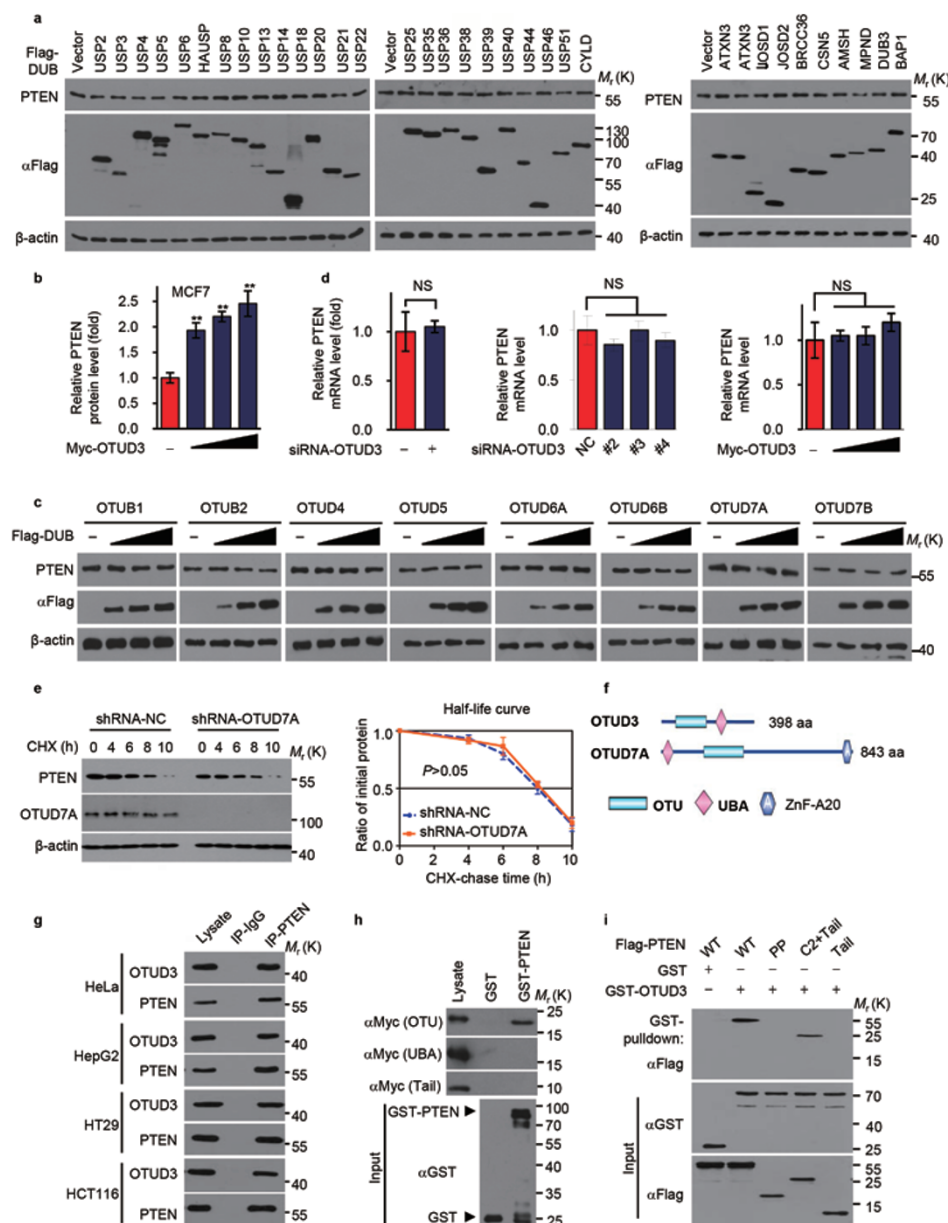
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Deubiquitylase OTUD3 regulates PTEN stability and suppresses tumorigenesis

Lin Yuan, Yanrong Lv, Hongchang Li, Haidong Gao, Shanshan Song, Yuan Zhang, Guichun Xing, Xiangzhen Kong, Lijing Wang, Yang Li, Tao Zhou, Daming Gao, Zhi-Xiong Xiao, Yuxin Yin, Wenyi Wei, Fuchu He and Lingqiang Zhang

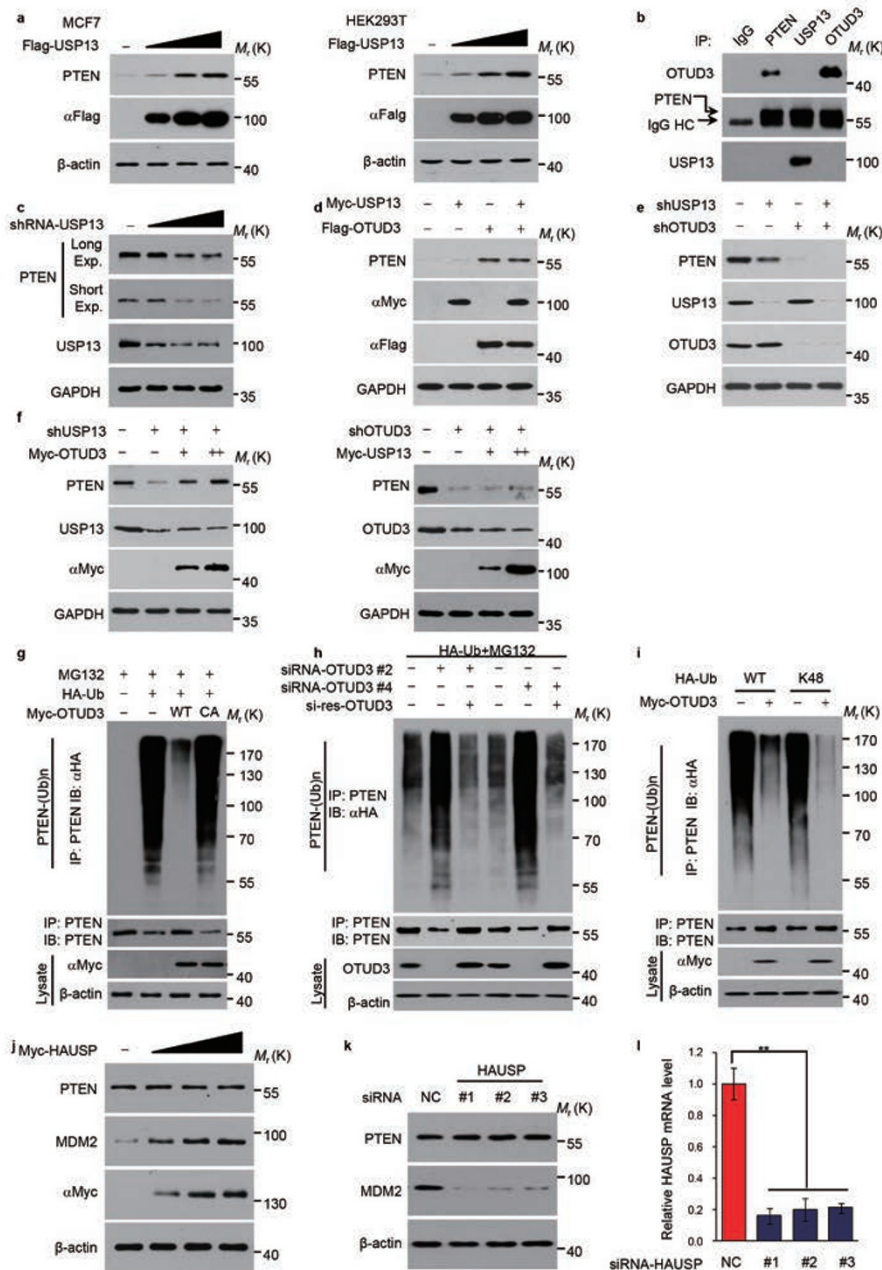
In the original version of this file the figures appeared in the wrong order. This error was corrected on 18 August 2015.

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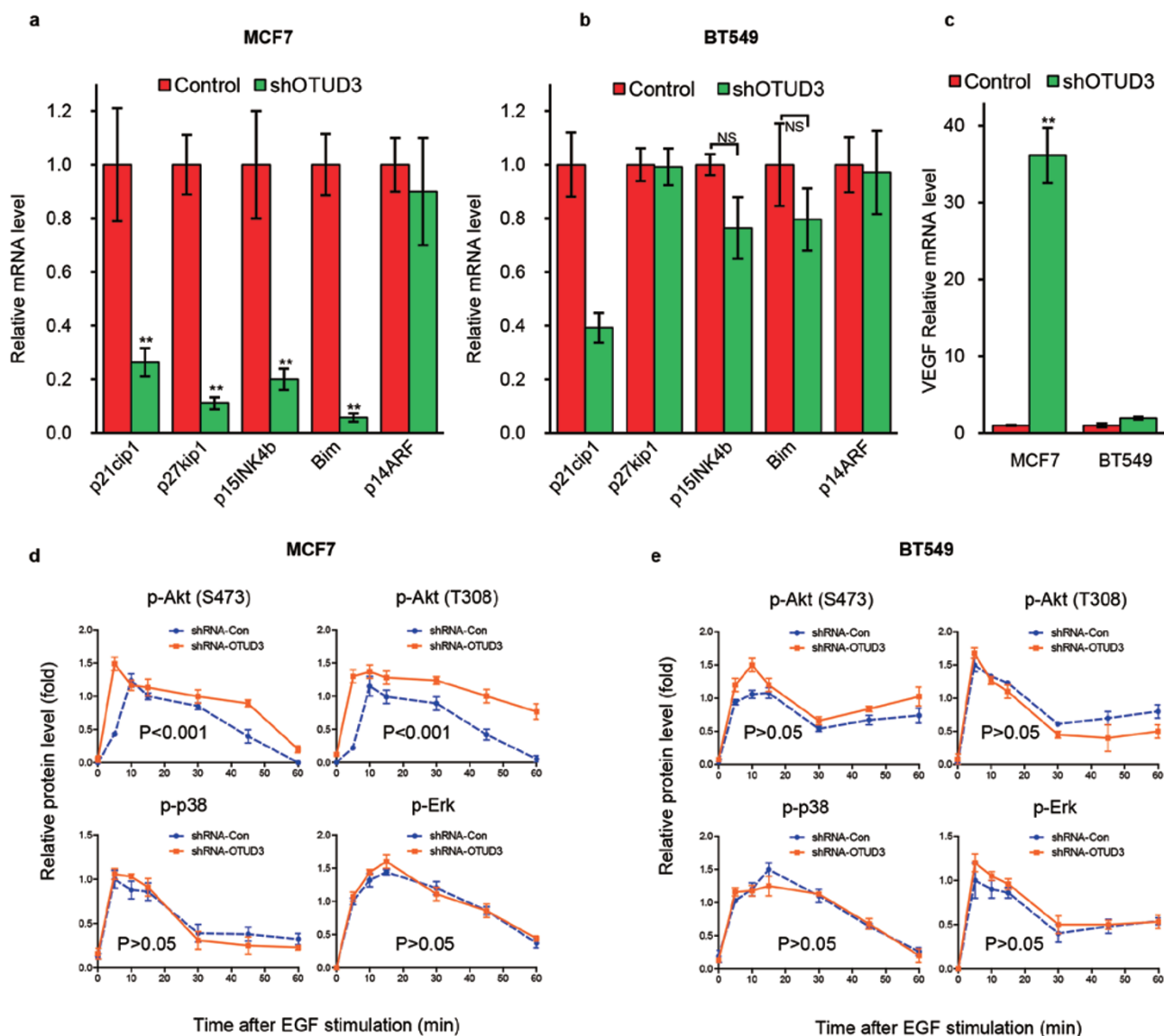
Supplementary Figure 1 OTUD3 stabilizes PTEN protein. **a**, Protein level analysis of PTEN in the presence or absence of overexpressed DUBs. HCT116 colon cancer cells were overexpressed with each of the Flag-tagged DUBs. **b**, Quantification of the PTEN protein levels relative to β -actin for Fig. 1b. $n=3$ independent experiments (**: $p < 0.01$, Student's t -test). **c**, Increasing amounts of plasmids encoding other members of OTU sub-family of DUBs were each transfected into HCT116 cells. The endogenous PTEN protein levels were examined by western blot. **d**, Knockdown of OTUD3 by pool siRNAs (left) or individual siRNAs (middle) or overexpression of OTUD3 (right) had no significant effect on PTEN mRNA level in MCF7 cells. $n=3$ independent experiments (NS: no significance, Student's t -test). **e**, MCF7 cells stably expressing the indicated shRNA were treated with cycloheximide (10 μ g ml⁻¹), and harvested at the indicated times. The left panels show

immunoblots of PTEN and OTUD7A. Right panel: quantification of the PTEN protein levels relative to β -actin. $n=3$ independent experiments (NS: no significance, two-way ANOVA test). **f**, Domain structure of OTUD3 and OTUD7A. **g**, Co-immunoprecipitation assays in four types of cancer cell lines to detect the endogenous interactions. The total cell lysates were immunoprecipitated with anti-OTUD3, anti-PTEN or control IgG as indicated. Both the lysates and the immunoprecipitates were subjected to western blot analysis. **h**, GST pull-down assays were performed to map the domain of OTUD3 required for interaction with PTEN. **i**, GST pull-down assays were performed to map the domain of PTEN required for interaction with OTUD3. All panels are representative results of three or more independent experiments. Statistics source data can be found in Supplementary Table 4. Uncropped images of blots are shown in Supplementary Fig. 9.



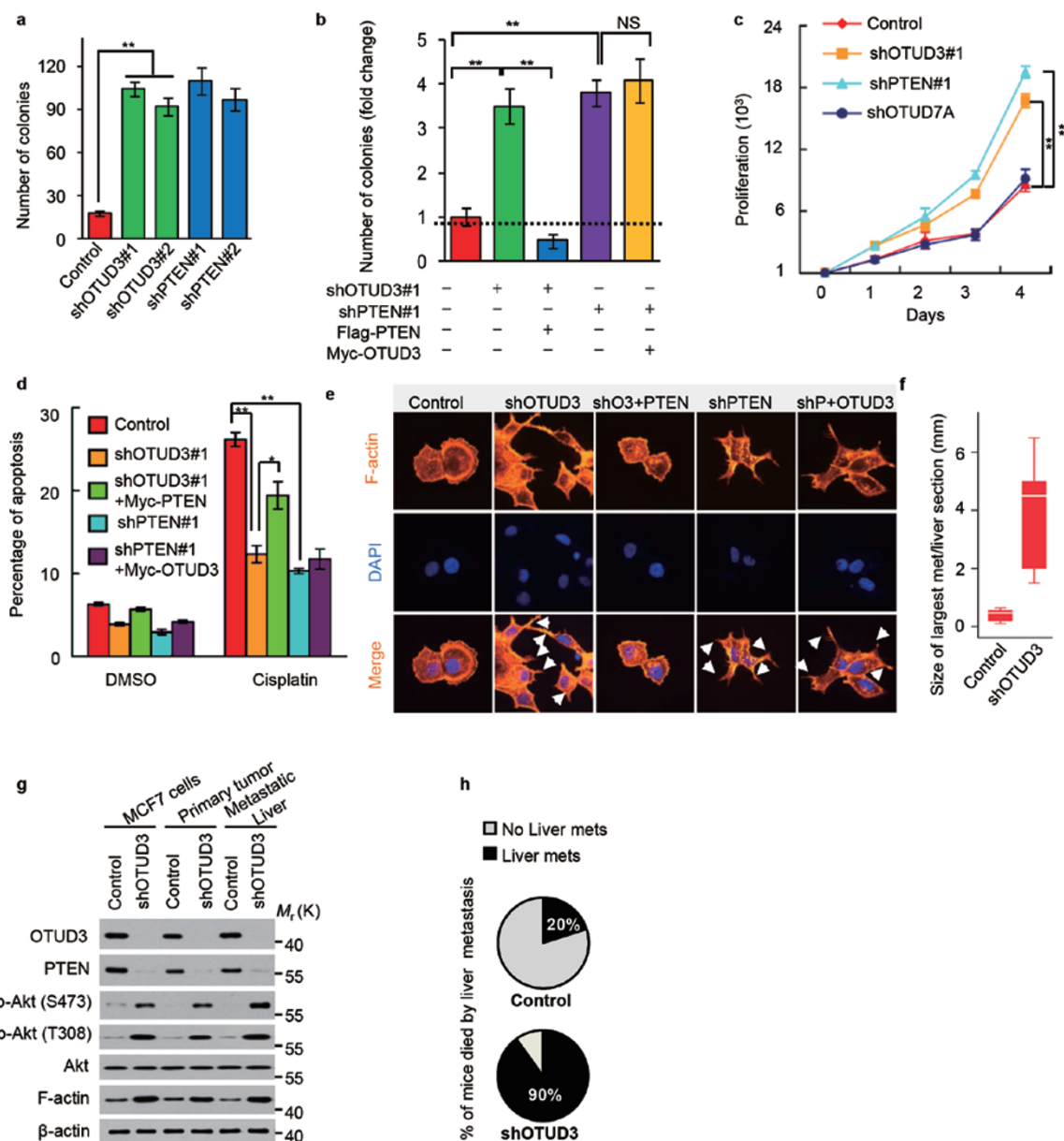
Supplementary Figure 2 OTUD3 is a more potent regulator of PTEN than USP13, and HAUSP has no significant effect on PTEN protein level. **a**, Increasing amounts of Flag-USP13 were transfected into MCF7 (left) and HEK293T (right) cells. After 48 h, PTEN protein levels were detected. **b**, Co-immunoprecipitation of endogenous PTEN, USP13 and OTUD3 in MCF7 cells. Protein extracts were immunoprecipitated with antibody against PTEN, USP13 or OTUD3, followed by immunoblotting with antibodies indicated. **c**, Increasing amounts of shRNA-USP13 were transfected into MCF7 cells. After 72 h, PTEN protein levels were detected. **d**, Protein level analysis of PTEN in the presence of overexpressed USP13 or OTUD3. MCF7 cells were overexpressed with Myc-USP13 or Flag-OTUD3. **e**, Protein level analysis of PTEN in MCF7 cells stably expressing the indicated shRNA. **f**, MCF7 cells were transfected with a constant amount of shRNA-USP13 and increasing amounts of Myc-OTUD3 (left), or a constant amount of shRNA-OTUD3 and increasing amounts of Myc-USP13 (right). After 48 h, PTEN protein levels were detected. **g**, HCT116 cells transfected with

the indicated constructs were treated with MG132 for 8 h before harvest. PTEN was immunoprecipitated with anti-PTEN and immunoblotted with anti-HA. **h**, siRNA-resistant OTUD3 was introduced into the HCT116 cells together with the OTUD3 siRNA. PTEN ubiquitination was measured. **i**, The PTEN ubiquitination linkage was analyzed in HCT116 cells transfected with OTUD3 and the indicated ubiquitin WT or K48-only plasmids. **j**, Increasing amounts of HAUSP plasmids were transfected into the MCF7 cells and the lysates were subjected to Western blot analysis. **k**, Knockdown of HAUSP by individual siRNAs had no significant effect on PTEN protein level but downregulated MDM2 protein level (positive) in MCF7 cells. **l**, Effectiveness of HAUSP knockdown by individual siRNAs was measured by quantitative RT-PCR. Data are shown as mean $\pm$ s.d. of $n=3$ independent experiments. (**: $p < 0.01$, Student's t -test). All panels are representative results of three or more independent experiments. Statistics source data can be found in Supplementary Table 4. Uncropped images of blots are shown in Supplementary Fig. 9.



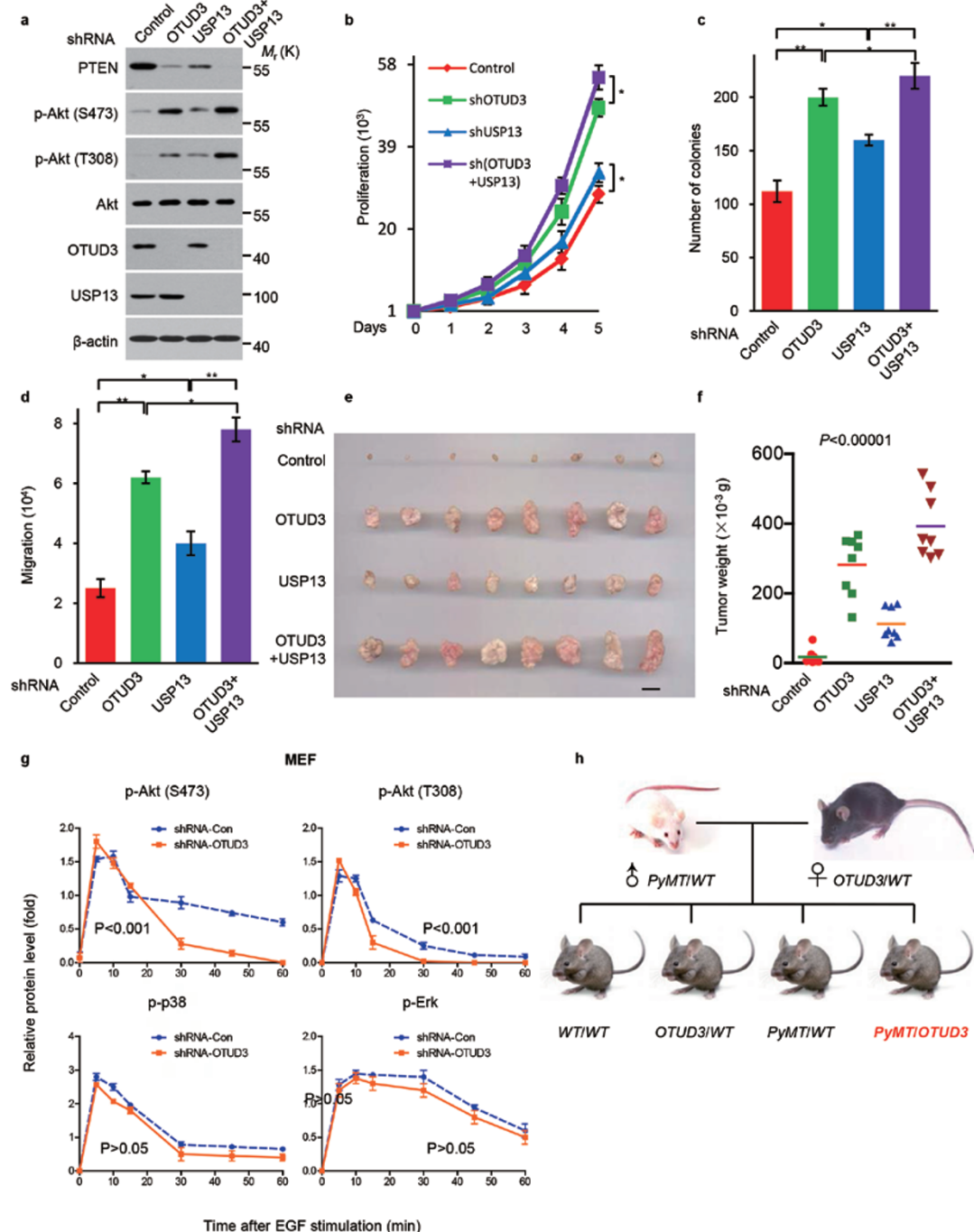
Supplementary Figure 3 OTUD3 negatively regulates Akt signaling in a PTEN-dependent manner. a-b, Quantitative RT-PCR analysis of indicated gene mRNA level in MCF7 (PTEN wild-type) (a) or BT549 (PTEN-null) (b) cells stably expressing the shRNA-Control or shOTUD3#1. Data were analyzed using two-tailed unpaired Student's *t*-test. Data are shown as mean $\pm$ s.d.. *n*=3 independent experiments. *: *p* < 0.05, **: *p* < 0.01, NS: no significance. c, Quantitative RT-PCR analysis of VEGF gene mRNA level in MCF7 and BT549 cells stably expressing shRNA-Control or shOTUD3 #1.

Data were analyzed using two-tailed unpaired Student's *t*-test. Data are shown as mean $\pm$ s.d.. *n*=3 independent experiments. d-e, OTUD3 shRNA-transduced MCF7 or BT549 cells were serum-starved and treated with EGF for various times. Quantification of the phosphorylation levels of Akt, p38 and Erk protein relative to total Akt, p38 and Erk protein levels for Fig. 3f and 3g. Data are shown as mean $\pm$ s.d.. *n*=3 independent experiments (two-way ANOVA test). Statistics source data can be found in Supplementary Table 4.



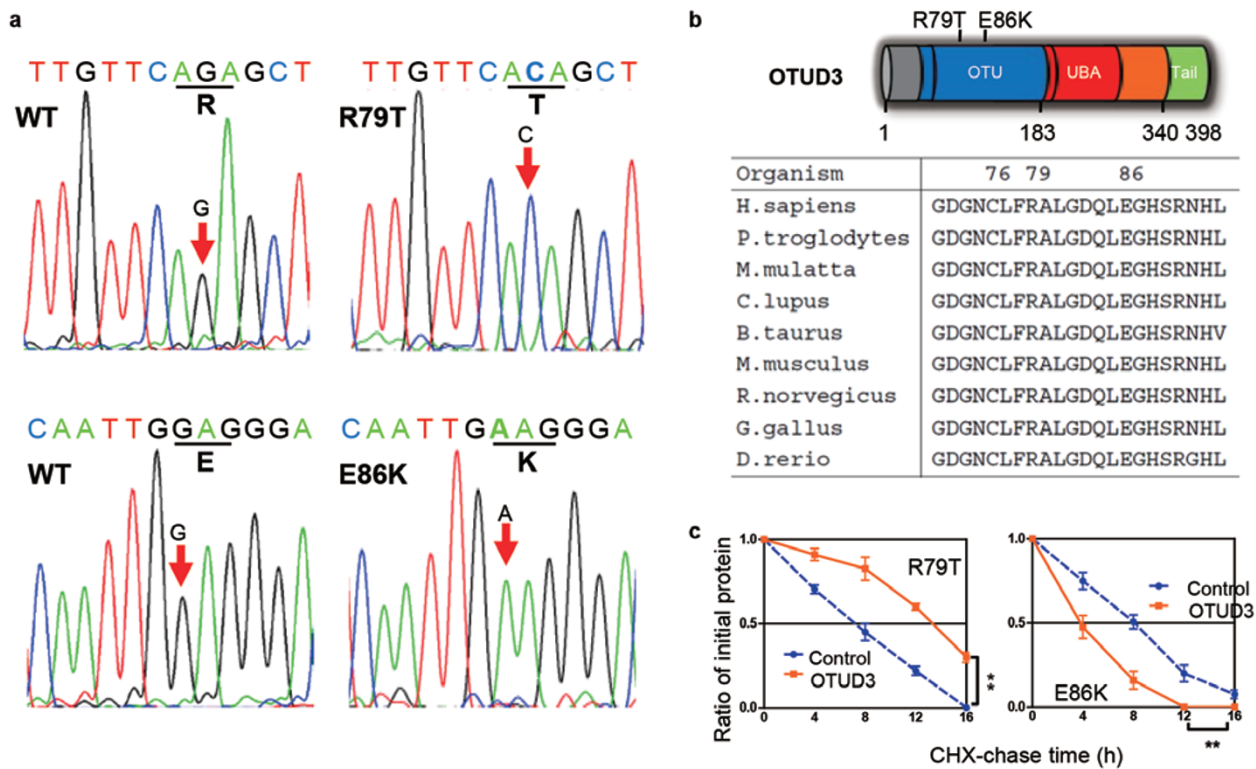
Supplementary Figure 4 OTUD3 inhibits cell proliferation, promotes cell apoptosis, and suppresses cell migration. **a**, Anchorage-independent growth of MCF7 cells stably expressing the indicated shRNA on soft agar. Viable colonies after 3 weeks were counted and the data from $n=3$ independent experiments were presented (mean $\pm$ s.d.). **: $p < 0.01$, Student's t -test. **b**, MCF7 cells stably expressing control shRNA, OTUD3 shRNA or PTEN shRNA were transfected with either OTUD3 or PTEN as indicated. Cells were tested for growth in colony assay. Colonies after one week were counted and the data (mean $\pm$ s.d.) from $n=3$ independent experiments were presented (**: $p < 0.01$, NS: no significance, Student's t -test, compared with cells expressing control shRNA;). **c**, MCF7 cells stably expressing control shRNA, OTUD3 shRNA or PTEN shRNA were seeded and cell proliferation was measured by MTS assay for 5 days. Data are shown as mean $\pm$ s.d.. $n=3$ independent experiments (**: $p < 0.01$, two-way ANOVA test). **d**, The indicated HCT116 cells were treated with cisplatin or DMSO to induce

apoptosis. The percentage of apoptotic cells was measured by Annexin-V staining. Data are shown as mean $\pm$ s.d.. $n=3$ independent experiments (**: $p < 0.05$; *: $p < 0.01$, Student's t -test). **e**, The cellular morphology was analyzed in MCF7 cells stably expressing the indicated shRNA and/or transfected with the OTUD3 or PTEN plasmids. Cells were fixed and stained for F-actin (red). Nuclei were stained with DAPI (blue). Scale bar, 20 μ m. **f**, Box plots showed the distribution of the size (largest in each section) of metastasis per section. Data were analyzed using Mann-Whitney test ($n=50$ fields from 10 independent mice, $p=0.0025$). **g**, Immunoblotting of OTUD3, PTEN, p-Akt, Akt, F-actin and β -actin in lysates of primary tumors and metastatic livers from mice injected with the stable shRNA-control or shRNA-OTUD3 MCF7 cells. **h**, Pie charts for mice died by liver metastasis. Data are representative results of 3 independent experiments. 3×10^3 Statistics source data can be found in Supplementary Table 4. Uncropped images of blots are shown in Supplementary Fig. 9.



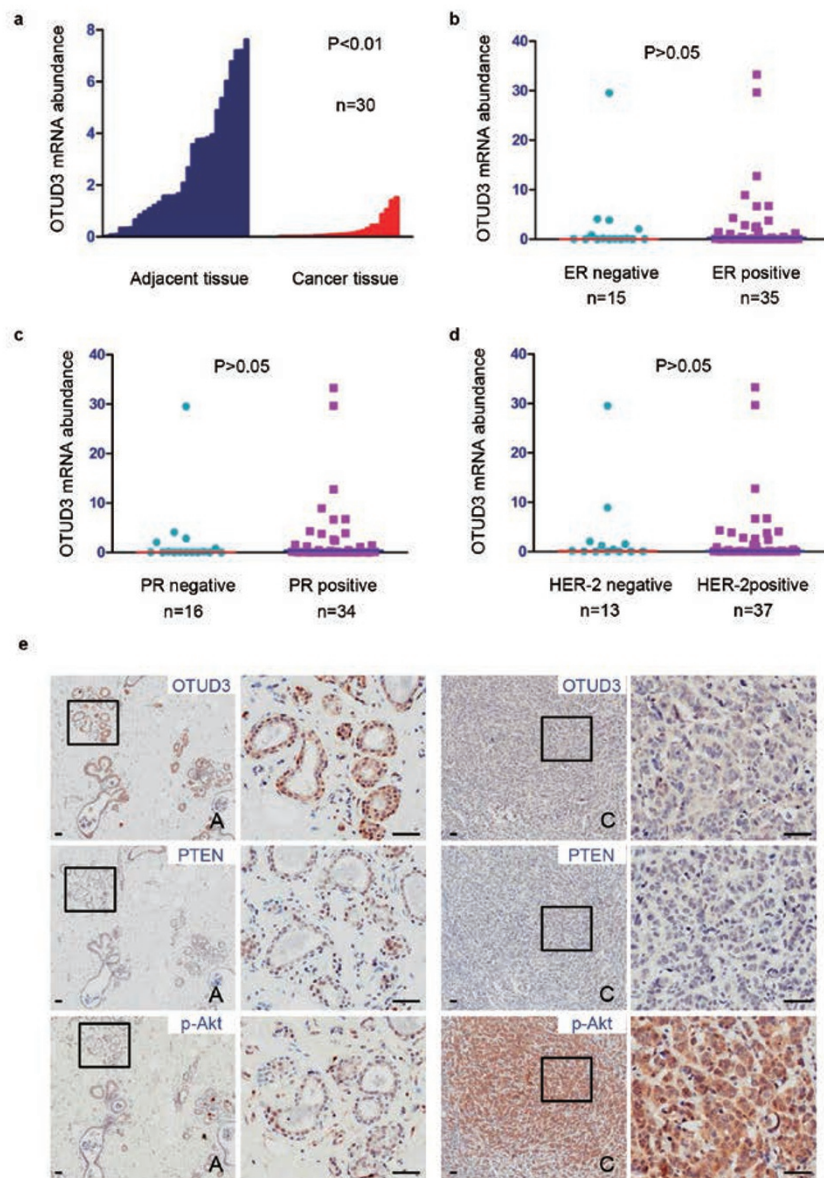
Supplementary Figure 5 The possible synergy between OTUD3 and USP13, and the phenotype of OTUD3 transgenic mice. **a**, Akt phosphorylation was analyzed in MCF7 cells stably expressing the indicated shRNA. **b–d**, The MCF7 cells expressing OTUD3 WT or mutants were measured for cell proliferation (**b**, two-way ANOVA test), anchorage-independent growth in soft agar (**c**, Student's *t*-test) and transwell migration assay (**d**, Student's *t*-test). Data are shown as mean $\pm$ s.d., $n=3$ independent experiments. **: $p < 0.01$, *: $p < 0.05$. **e**, Representative bright-field imaging of the tumors. MCF7 cells stably expressing indicated shRNA were implanted into a subcutaneous site in the skin on one flank of athymic nude mice ($n=8$). On 4 weeks, mice receiving transplants of indicated cells were sacrificed. Scale bar, 1 cm.

f, Whisker plots showed the distribution of the tumor weights ($n=8$ tumors from 8 independent mice). Data were shown as mean $\pm$ s.d. and analyzed using Kruskal-Wallis test. **g**, WT and TG MEFs were serum-starved and treated with EGF for various times. Quantification of the phosphorylation levels of Akt, p38 and Erk protein relative to total Akt, p38 and Erk protein levels for Fig. 5i. Data are shown as mean $\pm$ s.d., $n=3$ independent experiments (two-way ANOVA test). **h**, The mating strategy of OTUD3 TG mice crossing with MMTV-PyMT mice. Panels a,g are representative results of 3 or more independent experiments. 3×10^3 Statistics source data can be found in Supplementary Table 4. Uncropped images of blots are shown in Supplementary Fig. 9.



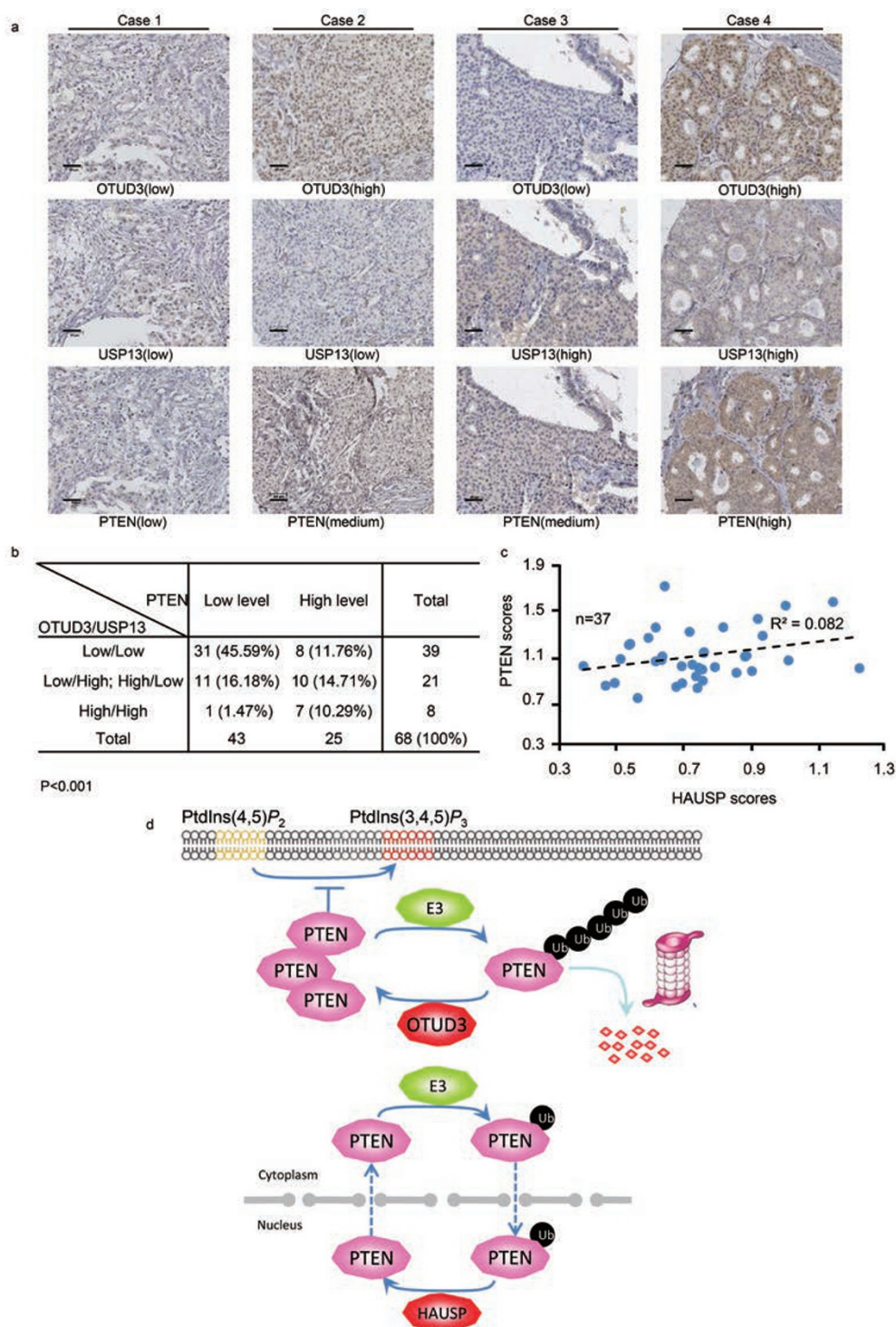
Supplementary Figure 6 OTUD3 E86K mutation in OTU domain leads to PTEN protein destabilization and increased tumorigenesis. **a**, Sequencing the region encoding OTUD3 OTU domain from 50 cases of Chinese breast cancer patients. Breast cancer tissues ($n=50$ independent samples) and the corresponding adjacent tissues ($n=50$ independent samples) were collected and subjected to RNA extraction and reverse transcription. The cDNAs were then sequenced. If the original sequencing results displayed heterozygous peaks, the corresponding cDNAs were ligated with pGEM-T vector (Promega)

and transformed into the competent bacteria. Then the colonies were sent out for sequencing. **b**, The mutation sites of OTUD3 in domain structure. Amino acid sequence alignment spanning OTUD3 R79 and E86 across species. **c**, Half-life of PTEN in the presence or absence of ectopic OTUD3 mutants. Quantification of the PTEN protein levels relative to β -actin for Fig. 6d (**: $p < 0.01$, two-way ANOVA test). Data are depicted as bar graphs with mean $\pm$ s.d.. $n=3$ independent experiments. Statistics source data can be found in Supplementary Table 4.



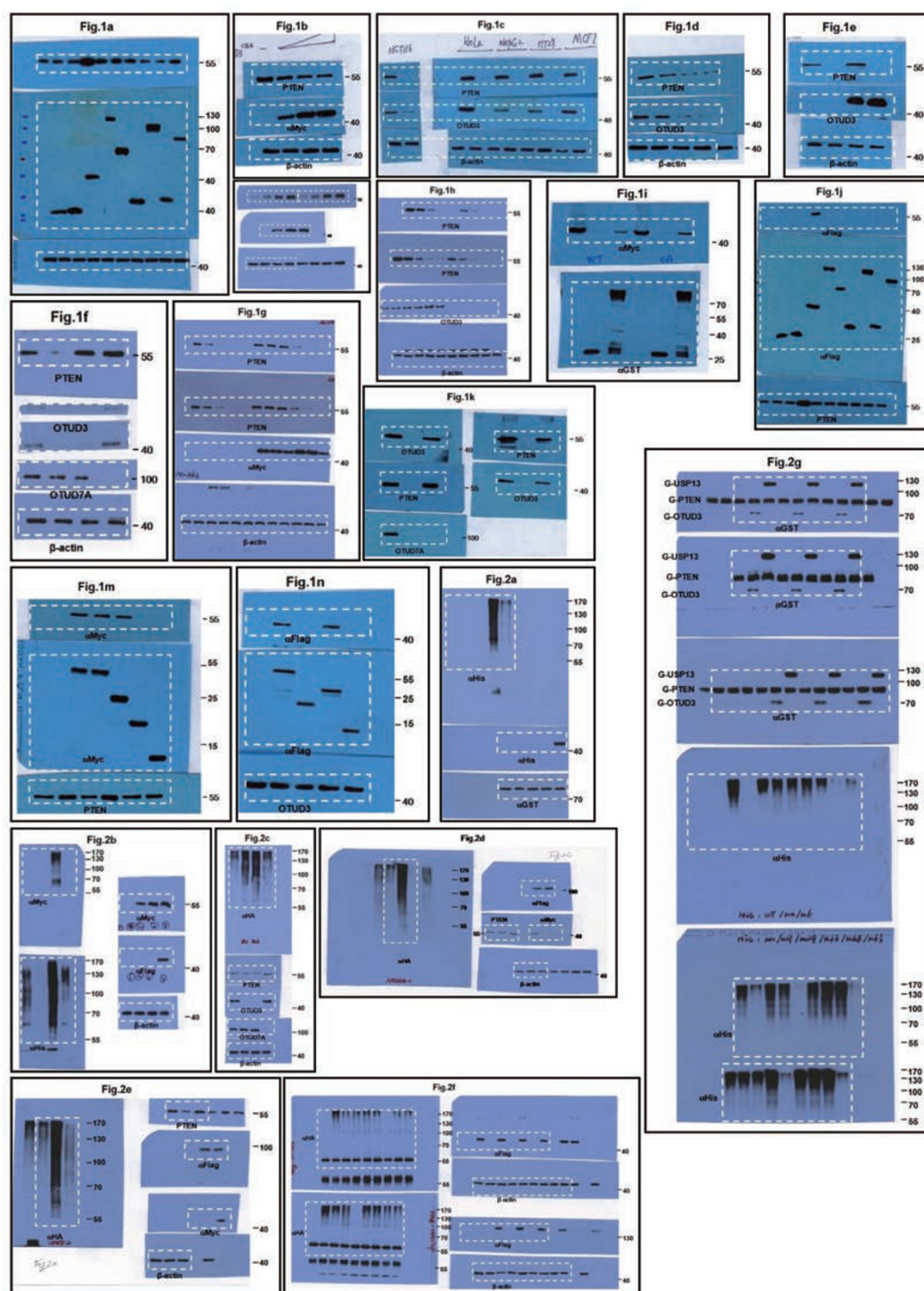
Supplementary Figure 7 OTUD3 expression is reduced in human breast cancer, which has no correlation with estrogen receptor, progesterone receptor or HER-2 expression. **a**, Relative OTUD3 mRNA level in breast cancer tissues and matched normal control from 30 subjects. Data were analyzed using Chi-square test. **b-d**, Whisker plots show the OTUD3 mRNA level in the cancer tissues classified into negative and positive groups based

on estrogen receptor (**b**), progesterone receptor (**c**) or HER-2 (**d**) expression from 50 subjects. Data were analyzed using Mann-Whitney test. **e**, Representative images from immunohistochemical staining of OTUD3, PTEN and p-Akt(S473) in three serial sections of the same tumor and matched adjacent tissue. The boxed areas in the left images were magnified on the right. A, adjacent tissue; C, carcinoma. Scale bar, 50 μm .

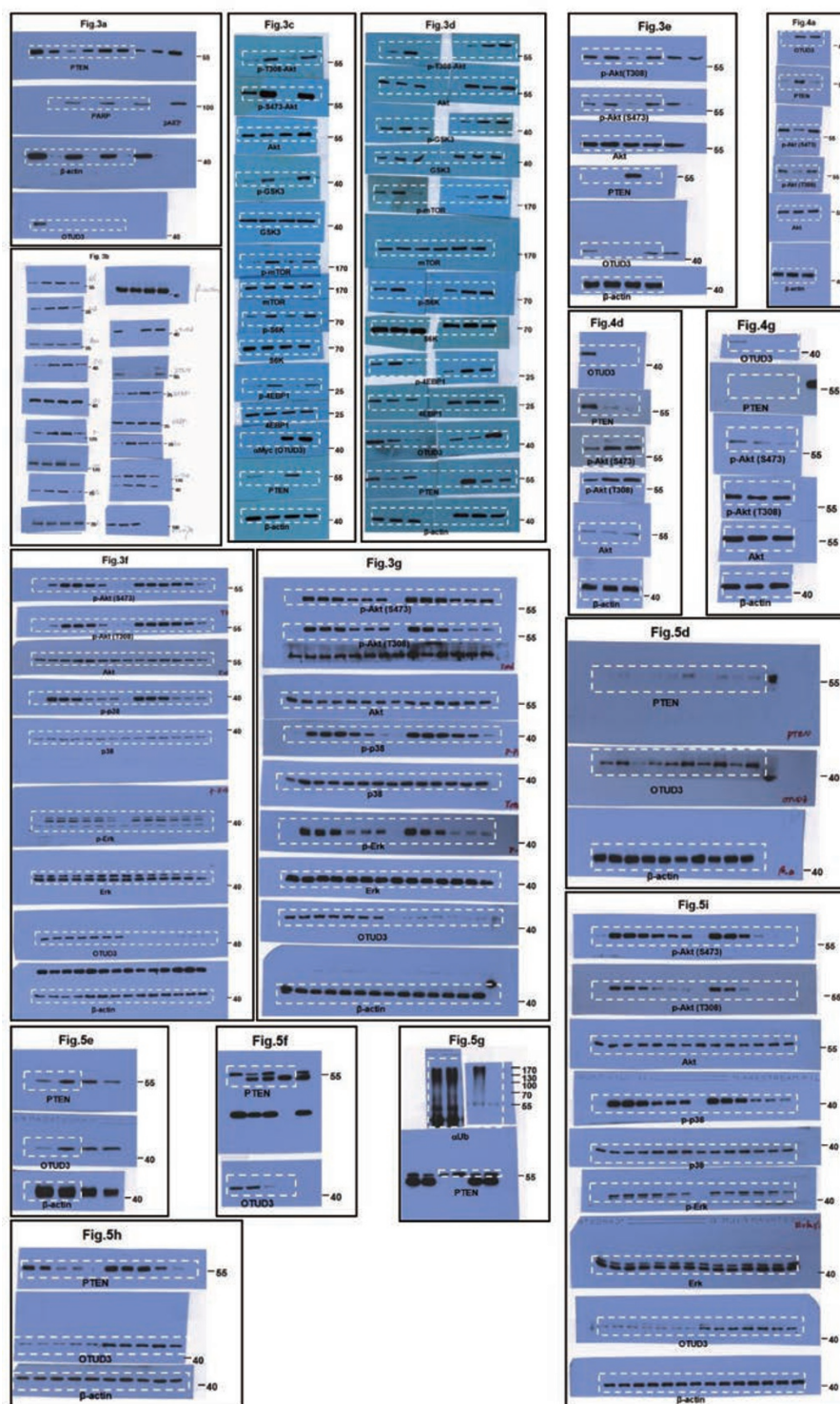


Supplementary Figure 8 OTUD3 and USP13 exhibit a synergy effect on PTEN expression in breast cancer samples. a,b, Potential synergy between OTUD3 and USP13 for PTEN protein levels. Serial sections of tissue arrays 68 patient breast specimens were subjected to immunohistochemistry with anti-OTUD3, anti-USP13 and anti-PTEN antibodies, respectively, and visualized by the DAB staining before imaging. Scale bar represents 50 μ m. Representative images were shown in (a), and the summary of the IHC results was listed in (b). $p < 0.001$, calculated by both Chi-square and χ^2 test. Scale bar, 50 μ m. c, Regression analysis comparing HAUSP and PTEN expression in breast cancer tissues. $n=37$. d, Proposed working model of OTUD3 on PTEN stability control.

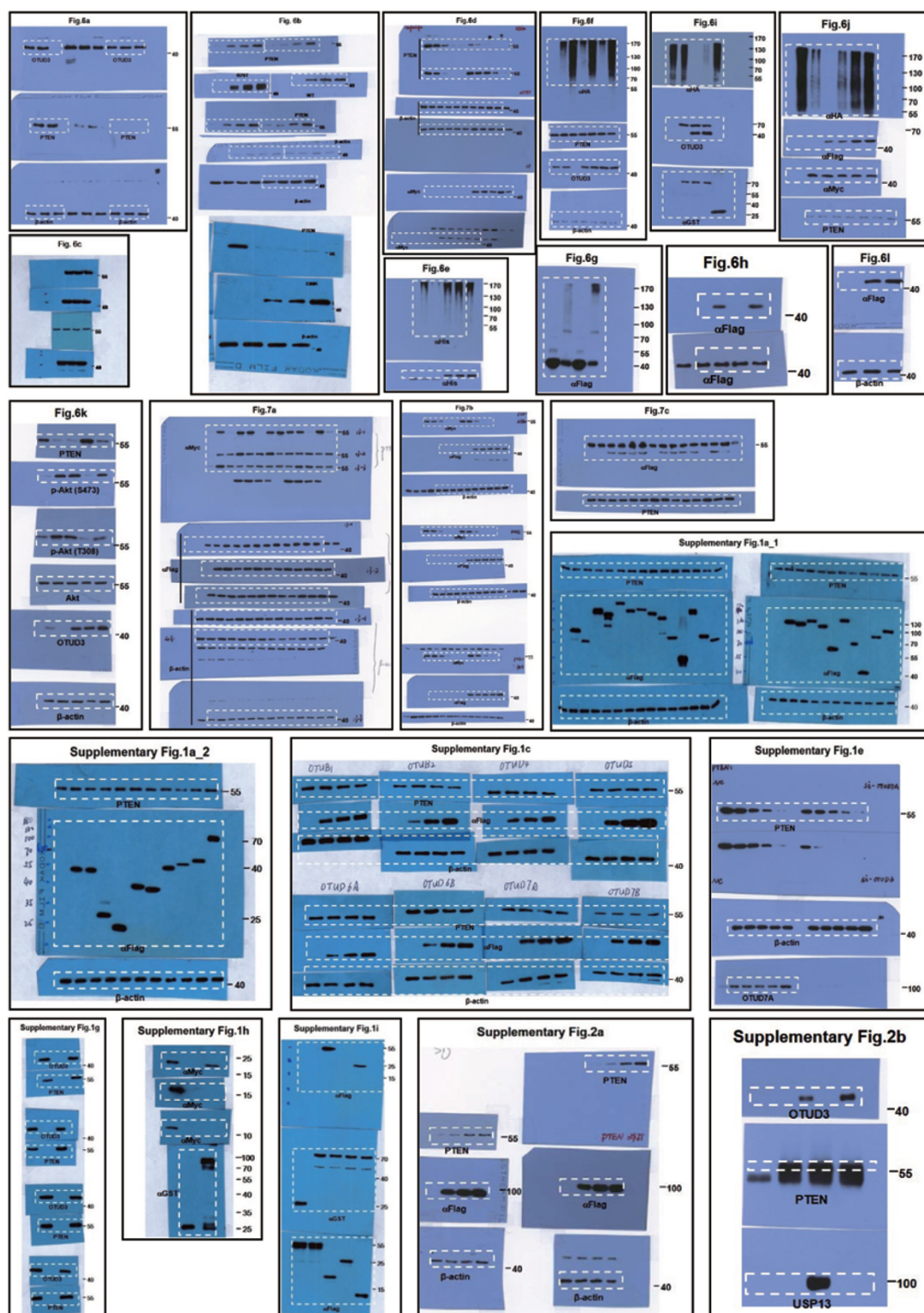
PTEN is a phosphatase that catalyzes the conversion of the lipid second messenger PtdIns(3,4,5)P3 to PtdIns(4,5)P2 and inhibits PI3K-Akt signaling. PTEN protein is relatively stable. This study identifies the deubiquitinase OTUD3 catalyzes the removal of poly-ubiquitin chain of PTEN in the cytoplasm. OTUD3 reverses the poly-ubiquitination of PTEN by E3 ubiquitin ligases (such as Nedd4-1, WWP2, XIAP and CHIP) and prevents PTEN degradation by 26S proteasome. On the other hand, a previous study (Ref. 22) identified that the deubiquitinase HAUSP is mainly localized in the nucleus and catalyzes the removal of mono-ubiquitination of PTEN. HAUSP regulates the PTEN cytoplasm-nucleus shuttling but has no effects on PTEN stability.



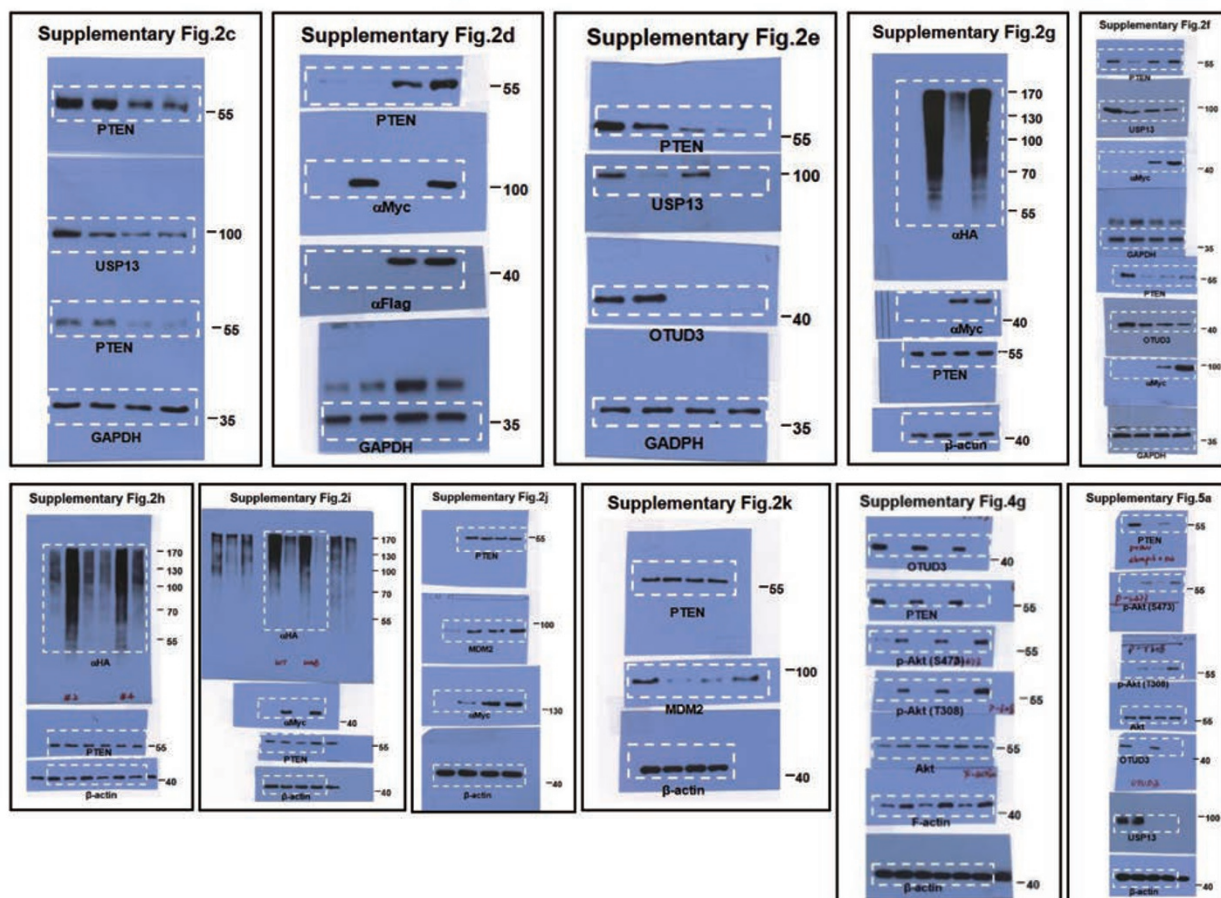
Supplementary Figure 9 Uncropped images of blots in Figures 1-7 and Supplementary Figures 1-5.



Supplementary Figure 9 continued



Supplementary Figure 9 continued



Supplementary Figure 9 continued

Table 1. OTUD3 mutations in human cancer. '+' shaded in light green represents the mutation can be queried in the databases.

	COSMIC	TCGA	CBIOPORTAL	CANCER
S89fs		+	+	Glioma
R90Q	+			Skin
K94R	+			Lung
R96K	+	+	+	Cutaneous Melanoma
R107L		+	+	Cutaneous Melanoma
E108V	+	+	+	Lung Adenocarcinoma
D109Y	+	+	+	Pancreas, Endometrial cancer
V114L	+			Skin
R144I	+	+	+	Endometrial cancer, Breast Invasive Carcinoma
A176T	+			Kidney
G180V	+	+	+	Kidney Renal Clear Cell Carcinoma
D184G			+	Hepatocellular Carcinom
D184N	+	+	+	Endometrial cancer
R187W	+	+	+	Stomach Adenocarcinoma, Lung Adenocarcinoma
D191E	+	+	+	Nasopharyngeal Carcinoma, Breast Invasive Carcinoma
L199P	+	+	+	Bladder Urothelial Carcinoma
D226V	+			Breast
D227N	+	+	+	Stomach Adenocarcinoma
N241N	+			Skin
R271W	+	+	+	Cutaneous Melanoma
R277I	+	+	+	NCI-60 Cell Lines, Stomach Adenocarcinoma
G302S			+	Pheochromocytoma and Paraganglioma
P327H	+	+	+	Stomach Adenocarcinoma
P327L	+	+	+	Esophageal Adenocarcinoma
V341I	+	+	+	Stomach Adenocarcinoma
T342fs		+	+	Cervical
Q350R			+	Hepatocellular Carcinom
Q358E	+			Thyroid
E359V	+			Thyroid
R361G	+			Thyroid
R363C	+	+	+	Colorectal Adenocarcinoma
E368G	+			Thyroid
S369R	+			Thyroid
R370S	+			Thyroid
N384S	+	+	+	Stomach Adenocarcinoma
607-2A>T	+			Kidney

Supplementary Table 1 OTUD3 mutations in human cancer. '+' shaded in light green represents the mutation can be queried in the databases.

Table 2. Primers for OTUD3 mutation selection in human breast cancer		
Gene (H.sapiens)	Forward	Reverse
OTUD3-Exon 1	TTACCTTCCCAACGCTTGAGG	GCCGCCCCGAATGAATGGATG
OTUD3-Exon 2	GATCTGGCATGAAGCATCATTG	ATGCCAGTGGACTTAATTACCC
OTUD3-Exon 3	TGGAACTCAGTAAATACCAAAGC	CCCCTCATCTTCGGATATTCC
OTUD3-Exon 4	GTGGGTGTTTCTCTCCAGAAG	TCAGGCCCCACAGATTCTCAATG
OTUD3-OTU (mRNA)	ATGTCCCGAAAGCAGGCGGCG	TTGATTTCATCTTGATGAAGCATCTG

Supplementary Table 2 Primers for OTUD3 mutation selection in human breast cancer.

Table 3. Primers for qPCR analysis

Gene (H.sapiens)	Forward	Reverse
OTUD3	TAAAGCAGCGGGAAGATTTTGA	TGCGATGTGTAACCTCCCTCAC
PTEN	TGGATTCTGACTTAGACTTGACCT	GGTGGGTATGGTCTTCAAAAGG
p21 ^{Cip1}	CGATGGAACTTCGACTTTGTCA	GCACAAGGGTACAAGACAGTG
p27 ^{Kip1}	AACGTGCGAGTGTCTAACGG	CCCTCTAGGGGTTTGTGATTCT
p15 ^{INK4b}	AGTCAACCGTTTCGGGAGGC	ACCACCAGCGTGTCCAGGAAG
VEGF	AGGGCAGAATCATCACGAAGT	AGGGTCTCGATTGGATGGCA
Bim	TAAGTTCTGAGTGTGACCGAGA	GCTCTGTCTGTAGGGAGGTAGG
p14ARF	GCCAGGGGCGCCCGCCGCTG	GGCCCGGTGCAGCACCACCA
p16	GCTGCCCAACGCACCGAATA	ACCACCAGCGTGTCCAGGAA
HAUSP	CCCTCCGTGTTTTGTGCGA	AGACCATGACGTGGAATCAGA
GAPDH	GGAGCGAGATCCCTCCAAAAT	GGCTGTTGTCATACTTCTCATGG
β-actin	CCTGGCACCCAGCACAAT	GCCGATCCACACGGAGTACTT

Supplementary Table 3 Primers for qPCR analysis.

Supplementary Table 4 The statistics source data for Figures and Supplementary Figures.